

Airline Satisfaction Analysis

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1. Introduction

Extenso relato N.1

2. Data Preparation

2.1 Data loading

2.2 Univariate Descriptive Analysis

In this section, we undertake a crucial step in any data science project, particularly for modeling purposes. A detailed descriptive analysis is performed on each variable, highlighting their most relevant characteristics. We follow a column-by-column examination of the dataset, and then generate a data quality report and prepare the data for the subsequent modeling stage by addressing missing values through imputation methods, and deleting non-informative variables and creating new ones.

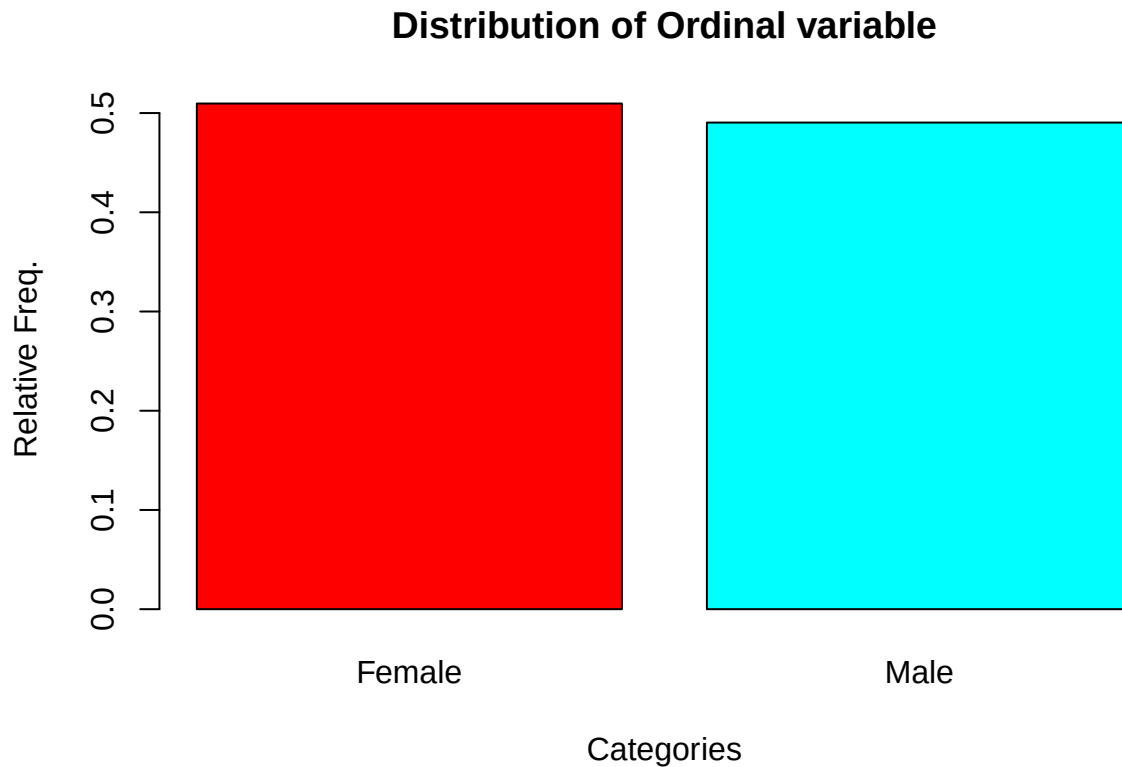
Note that the explanatory data analysis is assessed using exclusively the train data, even though new variables and variable removals also impact the test dataset. In this initial phase, we analyze all variables in the dataset, highlighting their most significant characteristics. We also convert numeric variables that represent qualitative concepts into factors and consider creating new variables based on existing numeric ones. Not only that but also ordinal numerical variables like ratings will be converted into ordered factors. Subsequently, new induced factor variables like age-brackets will also be introduced in this section.

2.2.1 Variable Gender

Gender is a categorical binary variable that represents the gender of the passenger which is almost perfectly balanced, with a few more females than males. There are no null values.

```
summarize_ordinal_data(train$Gender)
```

```
## [1] "Proportions: "  
## data  
## Female    Male  
## 0.5096 0.4904
```



```
## [1] "Mean: "  
## Warning in mean.default(data): argument is not numeric or logical: returning NA  
## [1] NA  
## [1] "Median: "  
## Warning in mean.default(sort(x, partial = half + 0L:1L)[half + 0L:1L]):  
## argument is not numeric or logical: returning NA  
## [1] NA  
## [1] "Standard Deviation: "  
## Warning in var(if (is.vector(x) || is.factor(x)) x else as.double(x), na.rm =  
## na.rm): NAs introduced by coercion  
## [1] NA  
## [1] "Number of nulls: "  
## [1] 0
```

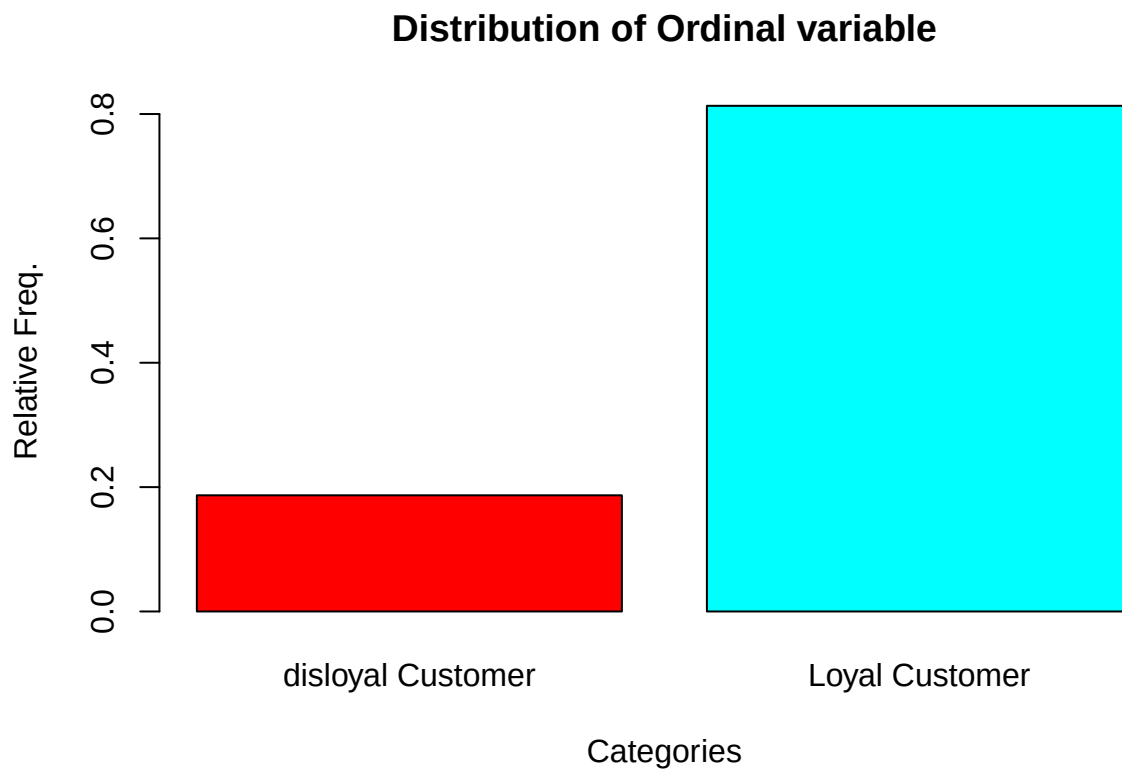
```
train$Gender <- as.factor(train$Gender)
test$Gender <- as.factor(test$Gender)
```

2.2.2 Variable Customer.Type

Customer.Type is a categorical binary variable that represents the kind of customer in terms of loyalty (Loyal or disloyal) which is unbalanced (81% vs 19%). There are no null values.

```
summarize_ordinal_data(train$Customer.Type)
```

```
## [1] "Proportions: "
## data
## disloyal Customer    Loyal Customer
##           0.1868           0.8132
```



```
## [1] "Mean: "
## Warning in mean.default(data): argument is not numeric or logical: returning NA
## [1] NA
## [1] "Median: "
## Warning in mean.default(sort(x, partial = half + 0L:1L)[half + 0L:1L]):
## argument is not numeric or logical: returning NA
## [1] NA
## [1] "Standard Deviation: "
## Warning in var(if (is.vector(x) || is.factor(x)) x else as.double(x), na.rm =
## na.rm): NAs introduced by coercion
```

```
## [1] NA
## [1] "Number of nulls: "
## [1] 0

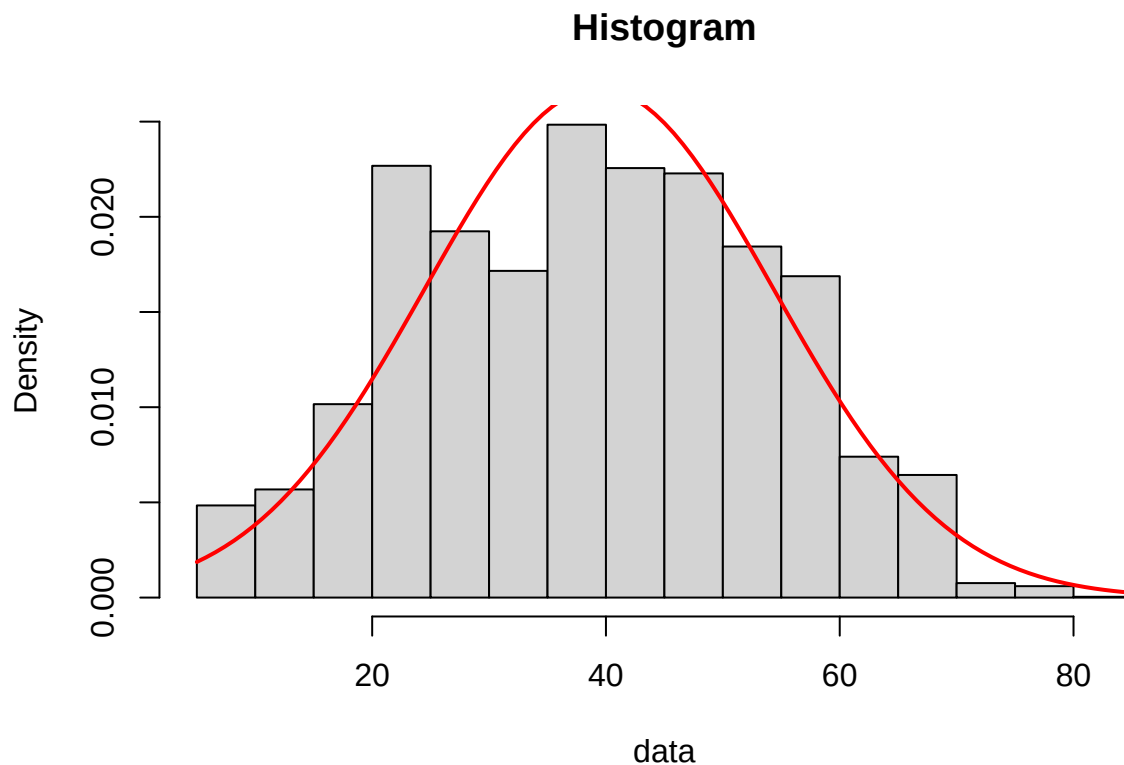
train$Customer.Type <- as.factor(train$Customer.Type)
test$Customer.Type <- as.factor(test$Customer.Type)
```

2.2.3 Variable Age

Age numerical variable represents the age of a given customer in the range [7, 85] and has a mean value of 39.41 with a standard deviation of 14.92. Considering the huge sample size and the lack of obvious normality, both inferential methods reject normality. Moreover, looking at the histogram and boxplot it doesn't show quite a normal behaviour. There are no null values nor outliers.

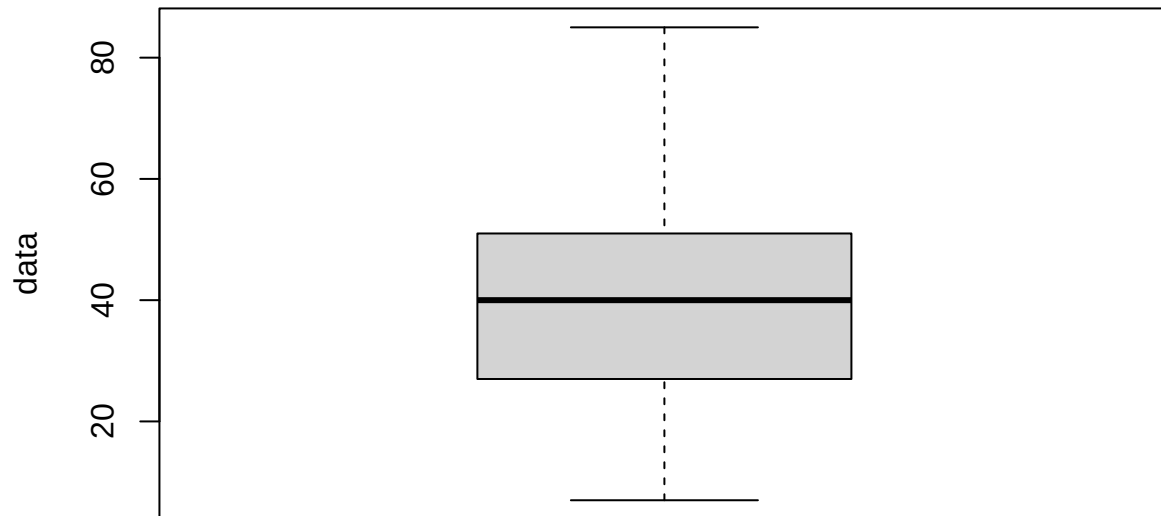
```
summary_and_normality_check(train$Age, bins=25)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      7.00  27.00   40.00   39.41   51.00   85.00
## [1] "Mean"
## [1] 39.4126
## [1] "Standard deviation"
## [1] 14.92196
```



```
## [1] "Skewness"
## [1] 0.0100164
```

Boxplot



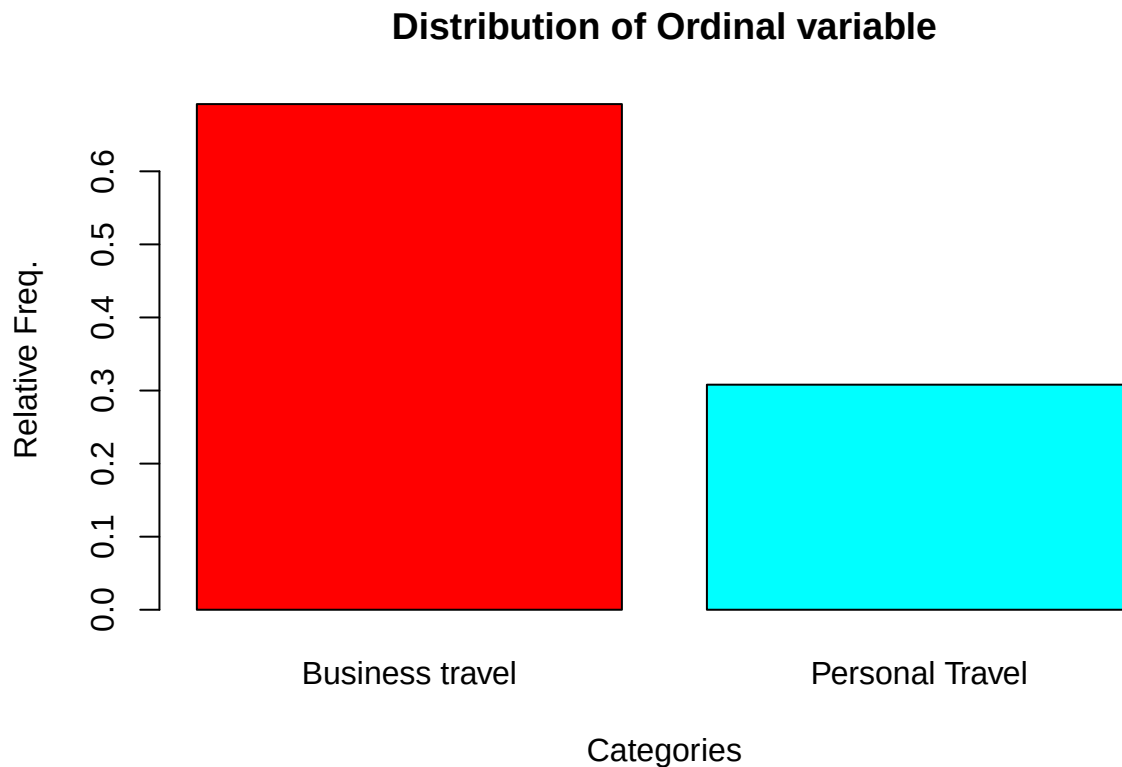
```
## [1] "Num Outliers:"
## [1] 0
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro-Wilk test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.98893, p-value < 2.2e-16
##
## [1] "Kolmogorov-Smirnov test: "
##
## Warning in ks.test.default(x = data, y = "pnorm", mean = mean(data), sd =
## sd(data)): ties should not be present for the one-sample Kolmogorov-Smirnov
## test
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: data
## D = 0.055434, p-value = 9.029e-14
## alternative hypothesis: two-sided
## integer(0)
```

2.2.4 Variable Type.of.Travel

Type.of.Travel categorical binary variable that represents the kind of flight of the observation (Business or Personal). The proportions are somewhat skewed towards business travel (0.692 vs 0.308) and there are no null values.

```
summarize_ordinal_data(train$Type.of.Travel)
```

```
## [1] "Proportions: "  
## data  
## Business travel Personal Travel  
##           0.692           0.308
```



```
## [1] "Mean: "  
## Warning in mean.default(data): argument is not numeric or logical: returning NA  
## [1] NA  
## [1] "Median: "  
## Warning in mean.default(sort(x, partial = half + 0L:1L)[half + 0L:1L]):  
## argument is not numeric or logical: returning NA  
## [1] NA  
## [1] "Standard Deviation: "  
## Warning in var(if (is.vector(x) || is.factor(x)) x else as.double(x), na.rm =  
## na.rm): NAs introduced by coercion  
## [1] NA  
## [1] "Number of nulls: "
```

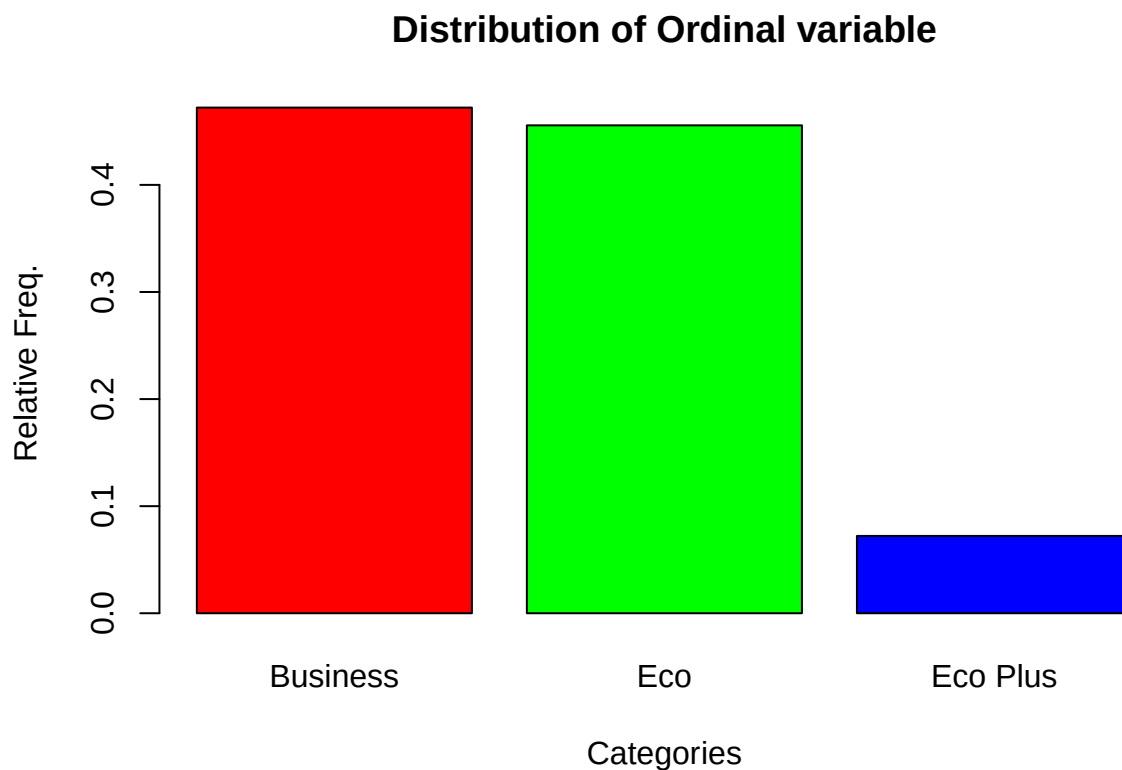
```
## [1] 0
train$Type.of.Travel <- as.factor(train$Type.of.Travel)
test$Type.of.Travel <- as.factor(test$Type.of.Travel)
```

2.2.5 Variable Class

Class categorical ternary variable that represents the type of cabin travel class of the observations (Business, Eco or Eco Plus). Vast majority of observations belong to Business or Eco evenly, and Eco plus has belongs to 7.22% of the observations. There are no null values.

```
summarize_ordinal_data(train$Class)
```

```
## [1] "Proportions: "
## data
## Business      Eco Eco Plus
##    0.4722    0.4556    0.0722
```



```
## [1] "Mean: "
## Warning in mean.default(data): argument is not numeric or logical: returning NA
## [1] NA
## [1] "Median: "
## Warning in mean.default(sort(x, partial = half + 0L:1L)[half + 0L:1L]):
## argument is not numeric or logical: returning NA
## [1] NA
## [1] "Standard Deviation: "
```

```
## Warning in var(if (is.vector(x) || is.factor(x)) x else as.double(x), na.rm =
## na.rm): NAs introduced by coercion

## [1] NA
## [1] "Number of nulls: "
## [1] 0

train$Class <- as.factor(train$Class)
test$Class <- as.factor(test$Class)
```

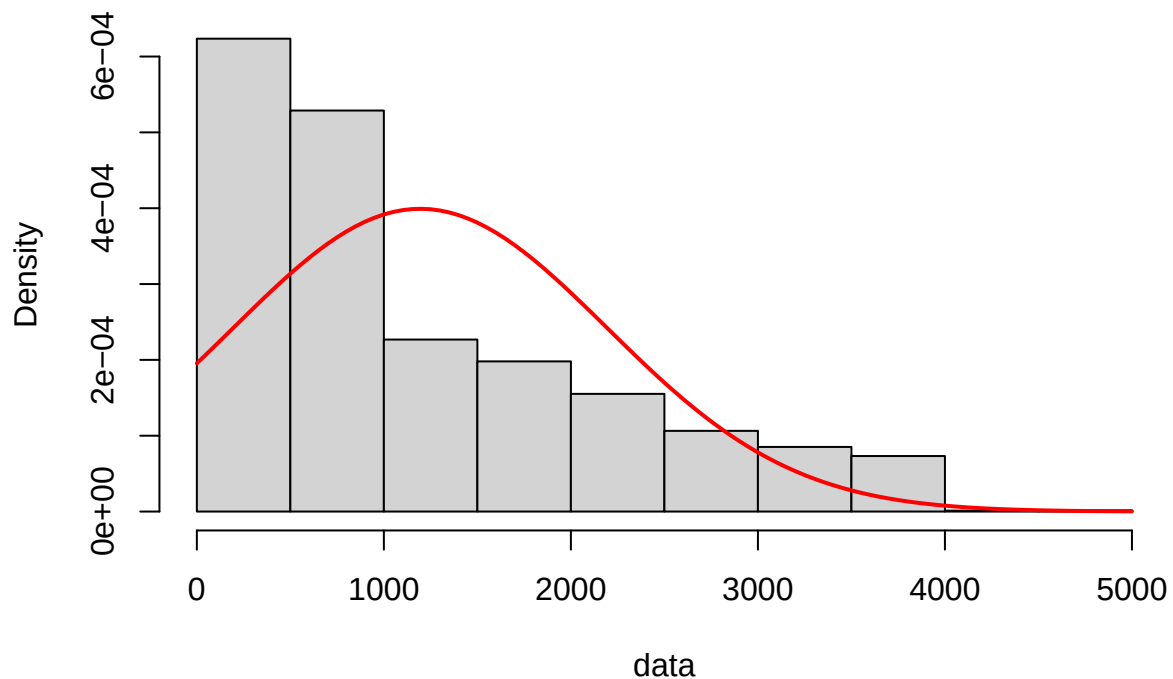
2.2.6 Variable Flight.Distance

Flight.Distance is a discrete finite numerical variable that represents the distance of the given flight in a range between [67, 4983] with a mean value of 1194 and 999 std. Its distribution is obviously not normal given the almost exponential shape that it has and both inferential tests reject with tiny p-values. There are 95 mild outliers and there is no metadata about the units.

```
summary_and_normality_check(train$Flight.Distance, bins=10)
```

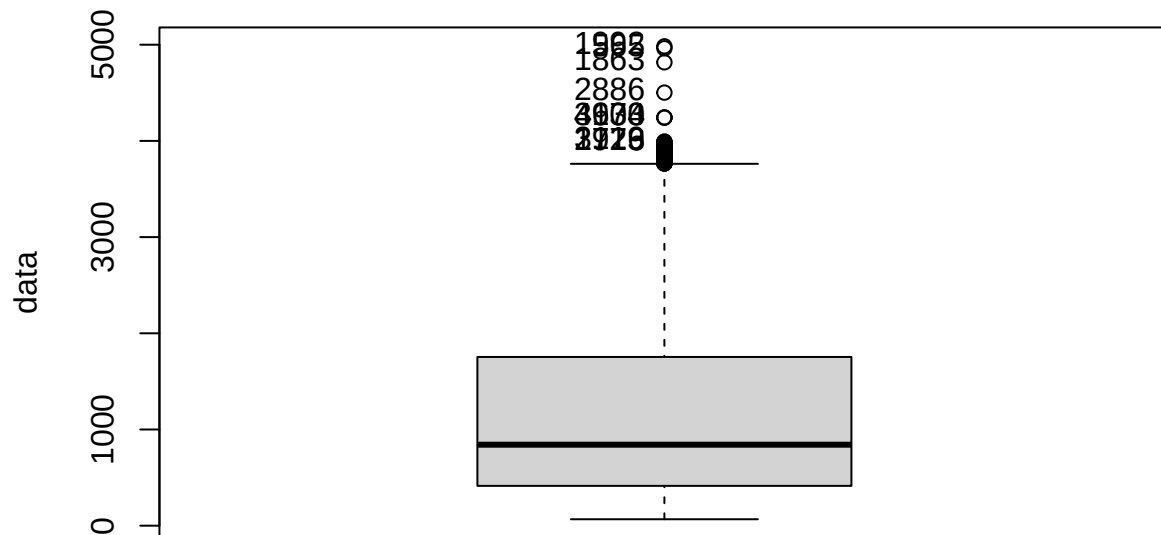
```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##       67    414     842   1194   1754   4983
## [1] "Mean"
## [1] 1194.205
## [1] "Standard deviation"
## [1] 999.2656
```

Histogram



```
## [1] "Skewness"
## [1] 1.087412
```

Boxplot



```
## [1] "Num Outliers:"
## [1] 95
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro-Wilk test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.86754, p-value < 2.2e-16
##
## [1] "Kolmogorov-Smirnov test: "
##
## Warning in ks.test.default(x = data, y = "pnorm", mean = mean(data), sd =
## sd(data)): ties should not be present for the one-sample Kolmogorov-Smirnov
## test
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: data
## D = 0.15624, p-value < 2.2e-16
## alternative hypothesis: two-sided
##
## [1] 205 206 221 379 388 458 523 533 565 622 692 720 825 878 880
## [16] 928 1090 1170 1272 1286 1428 1460 1470 1520 1579 1606 1758 1764 1825 1863
## [31] 1871 1887 1926 1992 2043 2101 2115 2295 2421 2474 2590 2624 2633 2720 2773
## [46] 2825 2859 2880 2886 2888 2974 2980 2985 3001 3119 3174 3215 3312 3333 3353
```

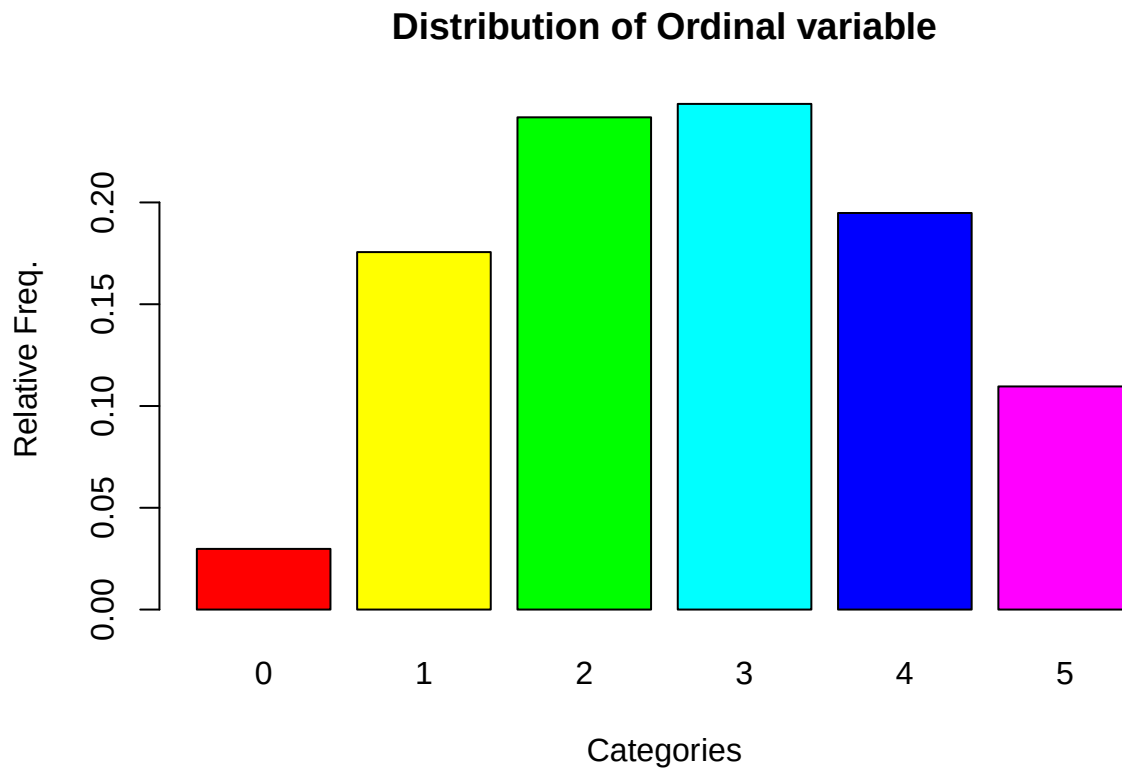
```
## [61] 3390 3533 3547 3617 3635 3680 3714 3744 3786 3821 3827 3837 3890 3916 3921
## [76] 3965 4020 4034 4183 4270 4278 4398 4553 4570 4587 4600 4639 4680 4689 4712
## [91] 4911 4918 4927 4933 4939
```

2.2.7 Variable Inflight.wifi.service

Inflight.wifi.service is a self-describing ordinal categorical variable given by the customers in the range [0,5]. The majority of values are grouped between 2 and 3, with a mean of 2.7, median of 3 and deviation of 1.33, producing a unimodal bell-shaped distribution centered at 2-3. There are no null values.

```
summarize_ordinal_data(train$Inflight.wifi.service)
```

```
## [1] "Proportions: "
## data
##      0      1      2      3      4      5
## 0.0298 0.1756 0.2418 0.2484 0.1948 0.1096
```



```
## [1] "Mean: "
## [1] 2.7316
## [1] "Median: "
## [1] 3
## [1] "Standard Deviation: "
## [1] 1.331884
## [1] "Number of nulls: "
## [1] 0
```

```
names(train)
```

```
## [1] "X" "id"
```

```
## [3] "Gender" "Customer.Type"
## [5] "Age" "Type.of.Travel"
## [7] "Class" "Flight.Distance"
## [9] "Inflight.wifi.service" "Departure.Arrival.time.convenient"
## [11] "Ease.of.Online.booking" "Gate.location"
## [13] "Food.and.drink" "Online.boarding"
## [15] "Seat.comfort" "Inflight.entertainment"
## [17] "On.board.service" "Leg.room.service"
## [19] "Baggage.handling" "Checkin.service"
## [21] "Inflight.service" "Cleanliness"
## [23] "Departure.Delay.in.Minutes" "Arrival.Delay.in.Minutes"
## [25] "satisfaction"

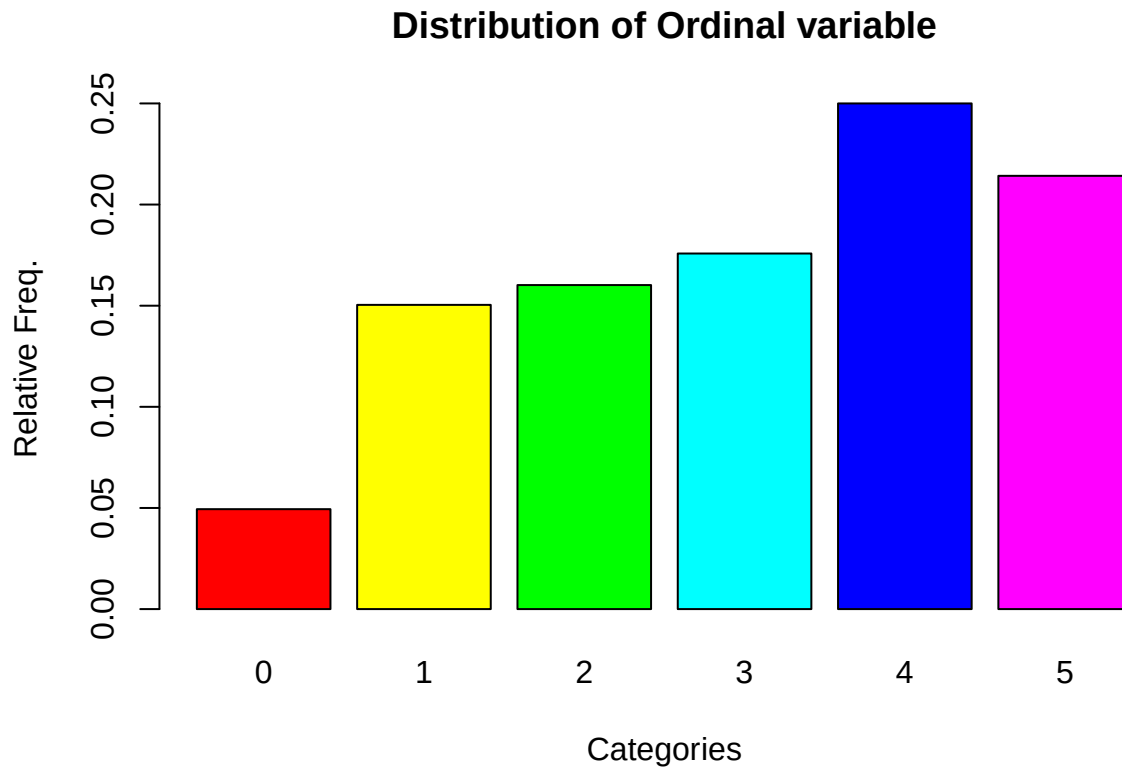
train$Inflight.wifi.service <- factor(train$Inflight.wifi.service,
                                     levels = c(0, 1, 2, 3, 4, 5),
                                     ordered = TRUE)
test$Inflight.wifi.service <- factor(test$Inflight.wifi.service,
                                     levels = c(0, 1, 2, 3, 4, 5),
                                     ordered = TRUE)
```

2.2.8 Variable Departure.Arrival.time.convenient

Departure.Arrival.time.convenient is a self-describing ordinal categorical variable given by the customers in the range [0,5]. The majority of values are grouped between 3 and 5, with a mean of 3, median of 3 and deviation of 1.51. The distribution is right-skewed and has a heavy left tail. There are no null values.

```
summarize_ordinal_data(train$Departure.Arrival.time.convenient)
```

```
## [1] "Proportions: "
## data
##      0      1      2      3      4      5
## 0.0494 0.1504 0.1602 0.1758 0.2500 0.2142
```



```
## [1] "Mean: "
## [1] 3.0692
## [1] "Median: "
## [1] 3
## [1] "Standard Deviation: "
## [1] 1.519498
## [1] "Number of nulls: "
## [1] 0

train$Departure.Arrival.time.convenient <- factor(train$Departure.Arrival.time.convenient,
  levels = c(0, 1, 2, 3, 4, 5),
  ordered = TRUE)

test$Departure.Arrival.time.convenient <- factor(test$Departure.Arrival.time.convenient,
  levels = c(0, 1, 2, 3, 4, 5),
  ordered = TRUE)
```

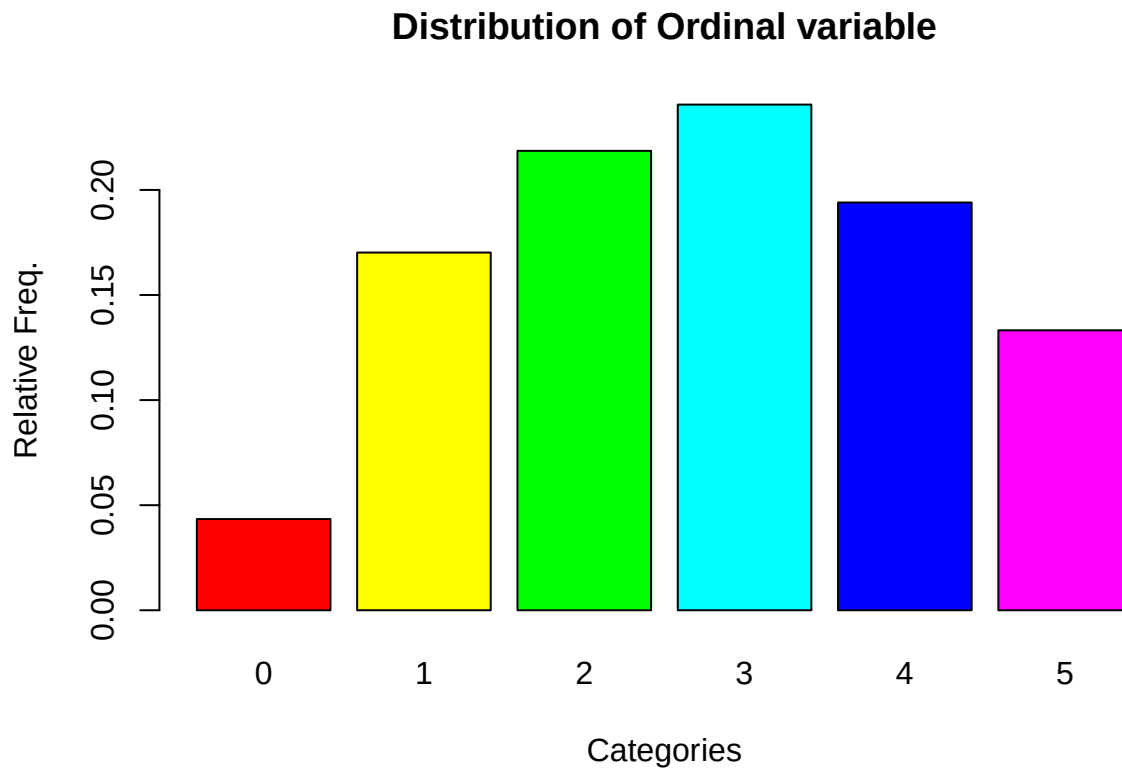
2.2.9 Variable Ease.of.Online.booking

Ease.of.Online.booking is a self-describing ordinal categorical variable given by the customers in the range [0,5]. The distribution of ratings is somewhat bell-shaped and unimodal, with a mean of 2.77, median 3 and deviation of 1.4. There are no null values.

```
summarize_ordinal_data(train$Ease.of.Online.booking)
```

```
## [1] "Proportions: "
## data
##      0      1      2      3      4      5
```

```
## 0.0434 0.1702 0.2186 0.2406 0.1940 0.1332
```



```
## [1] "Mean: "
## [1] 2.7712
## [1] "Median: "
## [1] 3
## [1] "Standard Deviation: "
## [1] 1.401729
## [1] "Number of nulls: "
## [1] 0

train$Ease.of.Online.booking <- factor(train$Ease.of.Online.booking,
                                       levels = c(0, 1, 2, 3, 4, 5),
                                       ordered = TRUE)

test$Ease.of.Online.booking <- factor(test$Ease.of.Online.booking,
                                       levels = c(0, 1, 2, 3, 4, 5),
                                       ordered = TRUE)
```

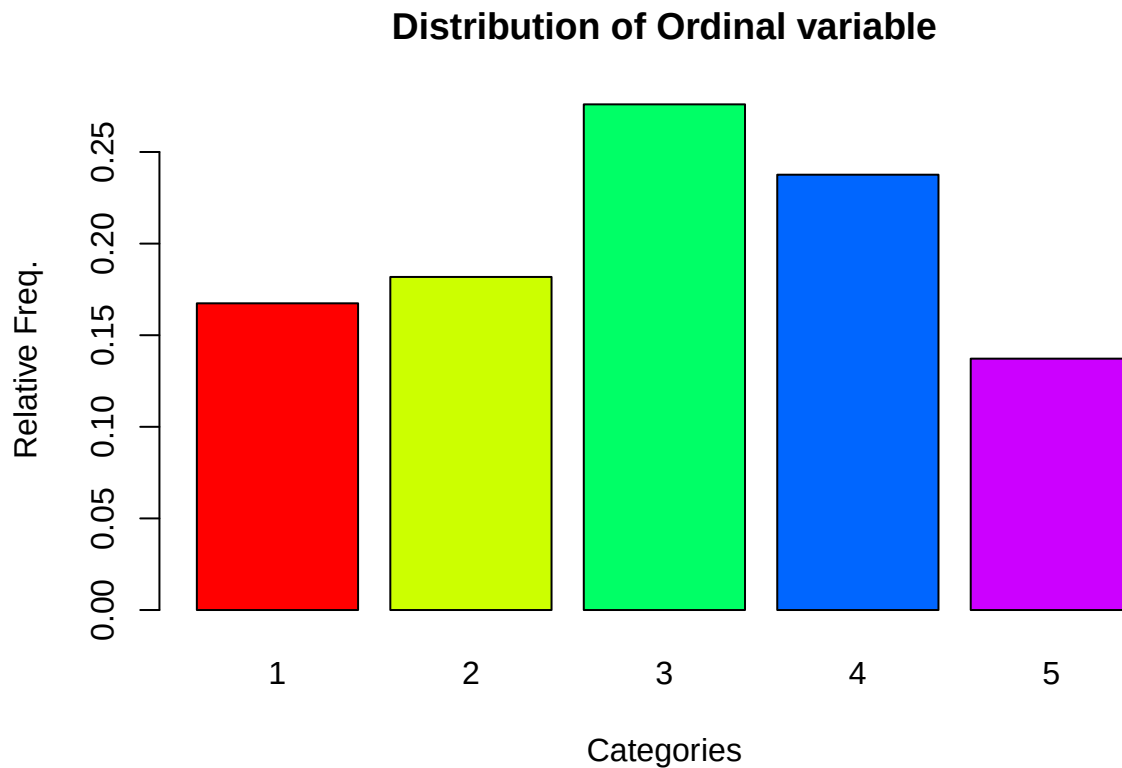
2.2.10 Variable Gate.location

Gate.location is a self-describing ordinal categorical variable given by the customers in the range [1,5]. The distribution is somewhat uniform with a median of 3, mean of 3 and deviation of 1.28. There are no null values.

```
summarize_ordinal_data(train$Gate.location)
```

```
## [1] "Proportions: "
```

```
## data
##      1      2      3      4      5
## 0.1674 0.1818 0.2760 0.2376 0.1372
```



```
## [1] "Mean: "
## [1] 2.9954
## [1] "Median: "
## [1] 3
## [1] "Standard Deviation: "
## [1] 1.279885
## [1] "Number of nulls: "
## [1] 0

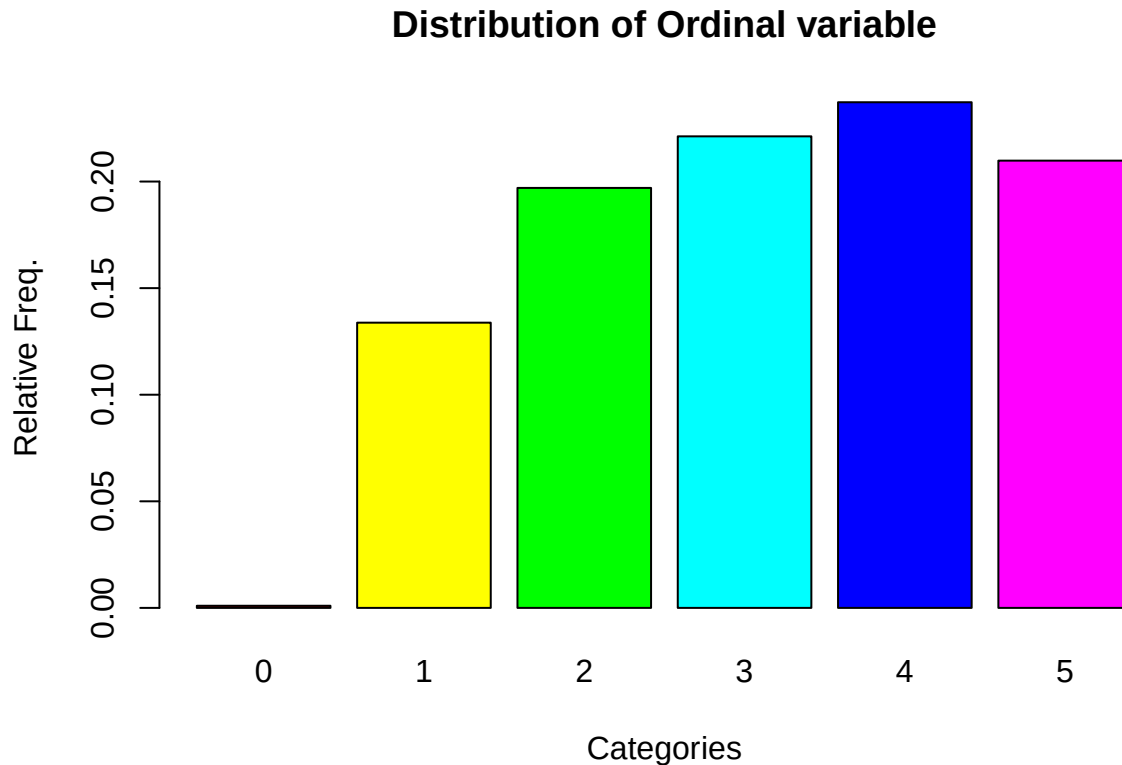
train$Gate.location <- factor(train$Gate.location,
                              levels = c(1, 2, 3, 4, 5),
                              ordered = TRUE)
test$Gate.location <- factor(test$Gate.location,
                              levels = c(1, 2, 3, 4, 5),
                              ordered = TRUE)
```

2.2.11 Variable Food.and.drink

Food.and.drink is a self-describing ordinal categorical variable given by the customers in the range [0,5]. The distribution is uniform and right skewed, with a median of 3, mean of 3.19 and deviation of 1.35. There are no null values.

```
summarize_ordinal_data(train$Food.and.drink)
```

```
## [1] "Proportions: "
## data
##      0      1      2      3      4      5
## 0.0010 0.1338 0.1970 0.2212 0.2372 0.2098
```



```
## [1] "Mean: "
## [1] 3.1892
## [1] "Median: "
## [1] 3
## [1] "Standard Deviation: "
## [1] 1.334976
## [1] "Number of nulls: "
## [1] 0

train$Food.and.drink <- factor(train$Food.and.drink,
                               levels = c(0, 1, 2, 3, 4, 5),
                               ordered = TRUE)

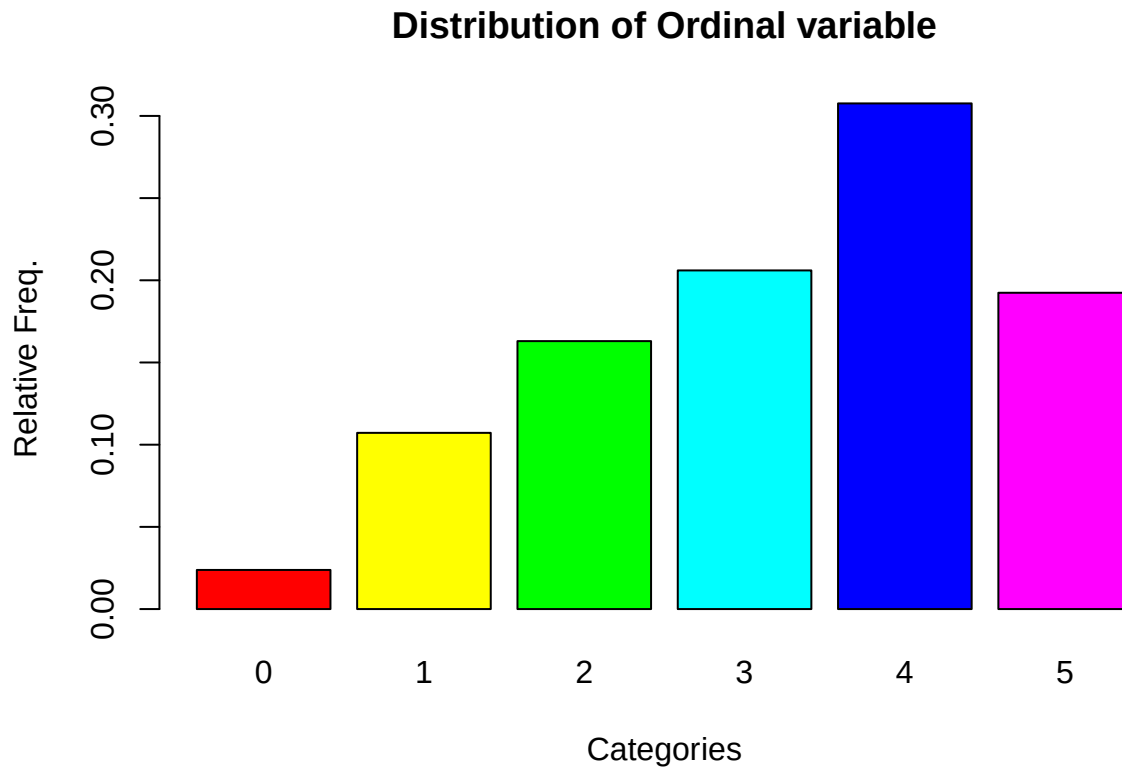
test$Food.and.drink <- factor(test$Food.and.drink,
                              levels = c(0, 1, 2, 3, 4, 5),
                              ordered = TRUE)
```

2.2.12 Variable Online.boarding

Online.boarding is a self-describing ordinal categorical variable given by the customers in the range [0,5]. The barplot showcases a fairly unimodal right skewed distribution, with a median of 3.5, mean of 3.24 and deviation of 1.35. There are no null values.

```
summarize_ordinal_data(train$Online.boarding)
```

```
## [1] "Proportions: "  
## data  
##      0      1      2      3      4      5  
## 0.0238 0.1072 0.1630 0.2060 0.3076 0.1924
```



```
## [1] "Mean: "  
## [1] 3.2436  
## [1] "Median: "  
## [1] 3.5  
## [1] "Standard Deviation: "  
## [1] 1.350638  
## [1] "Number of nulls: "  
## [1] 0
```

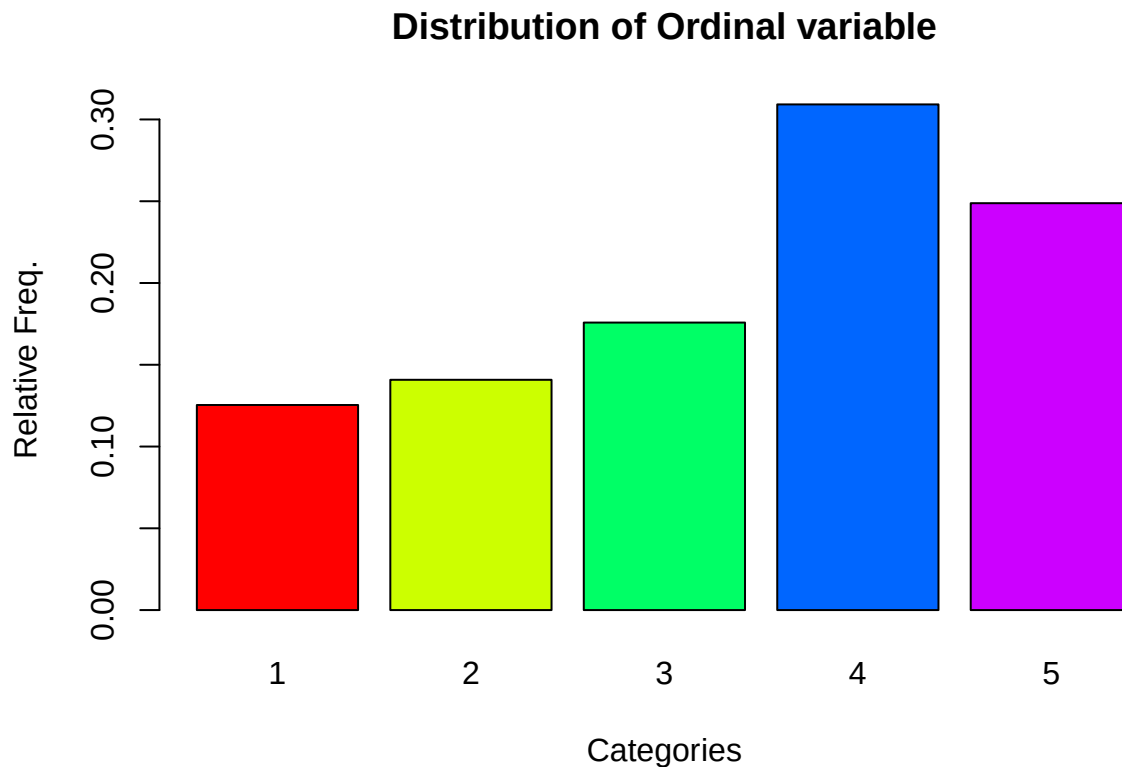
```
train$Online.boarding <- factor(train$Online.boarding,  
                                levels = c(0, 1, 2, 3, 4, 5),  
                                ordered = TRUE)  
  
test$Online.boarding <- factor(test$Online.boarding,  
                               levels = c(0, 1, 2, 3, 4, 5),  
                               ordered = TRUE)
```

2.2.12 Variable Seat.comfort

Seat.comfort is a self-describing ordinal categorical variable given by the customers in the range [1,5]. Its distribution is right skewed with a heavy left tail, and has a median of 4, mean of 3.41 and a deviation of 1.33. There are no null values.

```
summarize_ordinal_data(train$Seat.comfort)
```

```
## [1] "Proportions: "  
## data  
##      1      2      3      4      5  
## 0.1254 0.1408 0.1758 0.3092 0.2488
```



```
## [1] "Mean: "  
## [1] 3.4152  
## [1] "Median: "  
## [1] 4  
## [1] "Standard Deviation: "  
## [1] 1.332203  
## [1] "Number of nulls: "  
## [1] 0
```

```
train$Seat.comfort <- factor(train$Seat.comfort,  
                             levels = c(1, 2, 3, 4, 5),  
                             ordered = TRUE)  
  
test$Seat.comfort <- factor(test$Seat.comfort,  
                             levels = c(1, 2, 3, 4, 5),
```

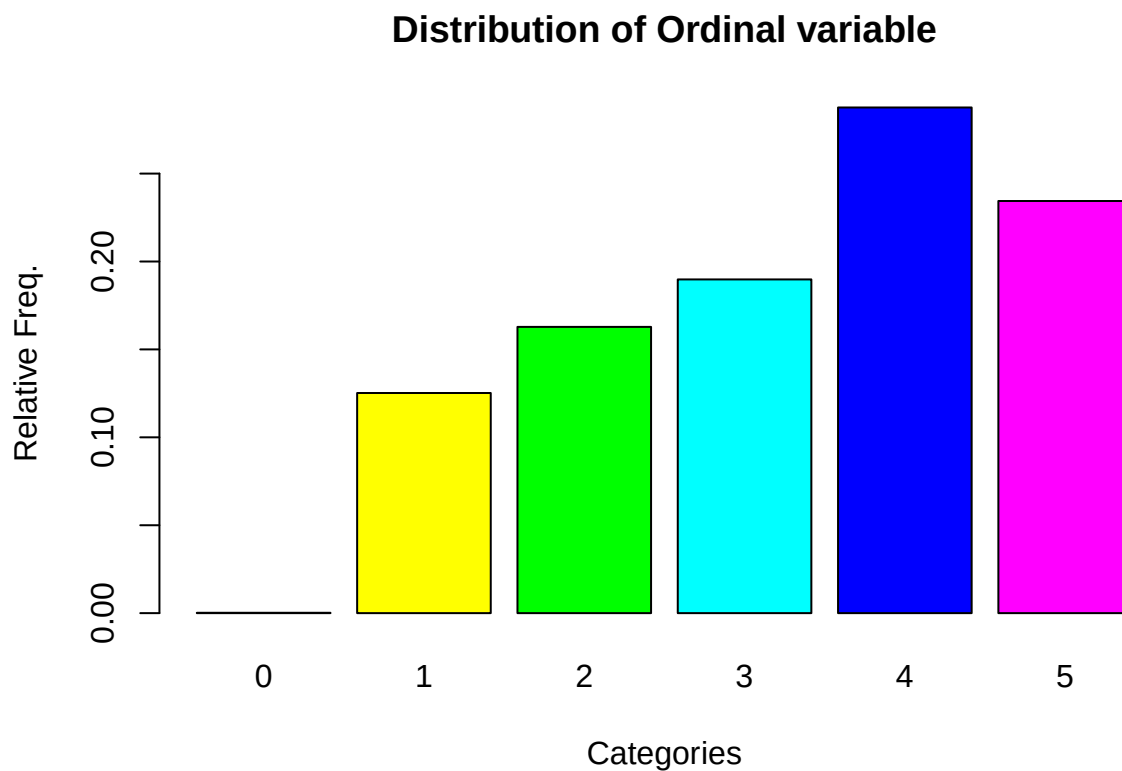
```
ordered = TRUE)
```

2.2.13 Variable Inflight.entertainment

Inflight.entertainment is a self-describing ordinal categorical variable given by the customers in the range [0,5]. Its distribution is unimodal and slightly right-skewed, with a median of 4, mean 3.34 and deviation of 1.33. There are no nulls

```
summarize_ordinal_data(train$Inflight.entertainment)
```

```
## [1] "Proportions: "  
## data  
##      0      1      2      3      4      5  
## 0.0002 0.1252 0.1628 0.1898 0.2876 0.2344
```



```
## [1] "Mean: "  
## [1] 3.3426  
## [1] "Median: "  
## [1] 4  
## [1] "Standard Deviation: "  
## [1] 1.331758  
## [1] "Number of nulls: "  
## [1] 0  
  
train$Seat.comfort <- factor(train$Seat.comfort,  
                             levels = c(0, 1, 2, 3, 4, 5),  
                             ordered = TRUE)
```

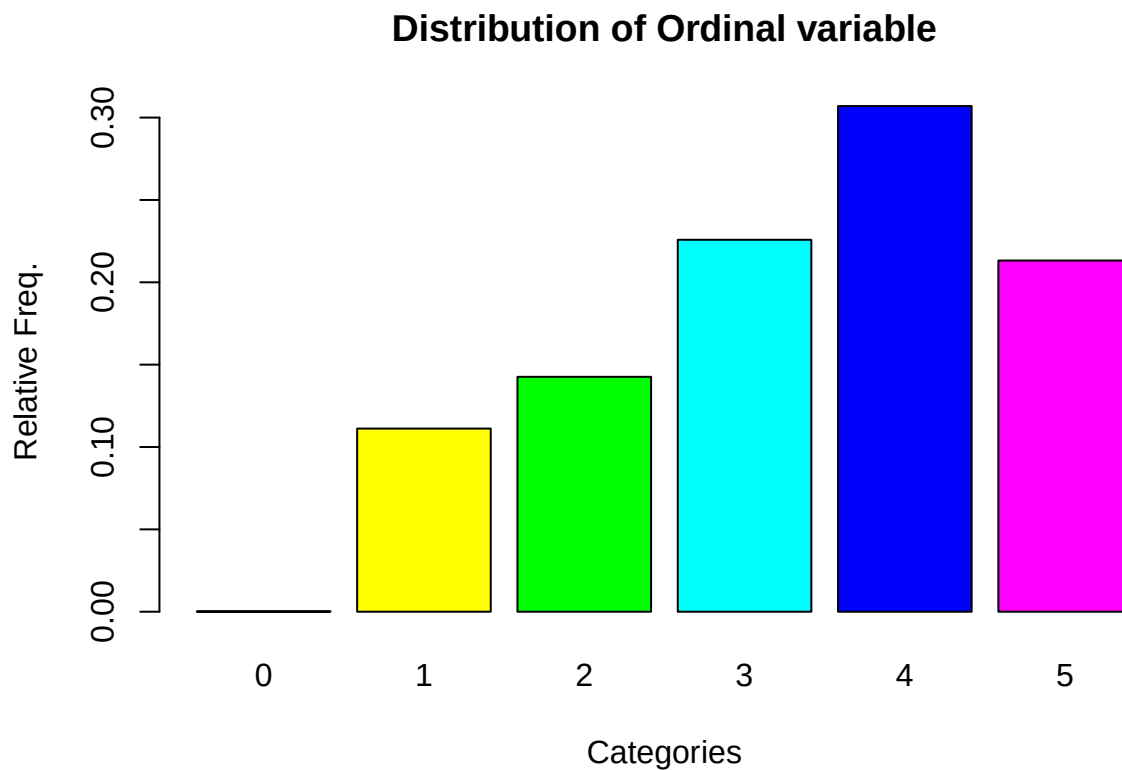
```
test$Seat.comfort <- factor(test$Seat.comfort,
                             levels = c(0, 1, 2, 3, 4, 5),
                             ordered = TRUE)
```

2.2.14 Variable On.board.service

On.board.service is a self-describing ordinal categorical variable given by the customers in the range [0,5]. Similarly, its distribution is unimodal and right skewed with a median of 4, mean of 3.37 and deviation of 1.27. There are no null values.

```
summarize_ordinal_data(train$On.board.service)
```

```
## [1] "Proportions: "
## data
##      0      1      2      3      4      5
## 0.0002 0.1112 0.1426 0.2258 0.3070 0.2132
```



```
## [1] "Mean: "
## [1] 3.3678
## [1] "Median: "
## [1] 4
## [1] "Standard Deviation: "
## [1] 1.270451
## [1] "Number of nulls: "
## [1] 0
```

```
train$On.board.service <- factor(train$On.board.service,
                                  levels = c(0, 1, 2, 3, 4, 5),
```

```

                                ordered = TRUE)

test$On.board.service <- factor(test$On.board.service,
                                levels = c(0, 1, 2, 3, 4, 5),
                                ordered = TRUE)

```

2.2.15 Variable Leg.room.service

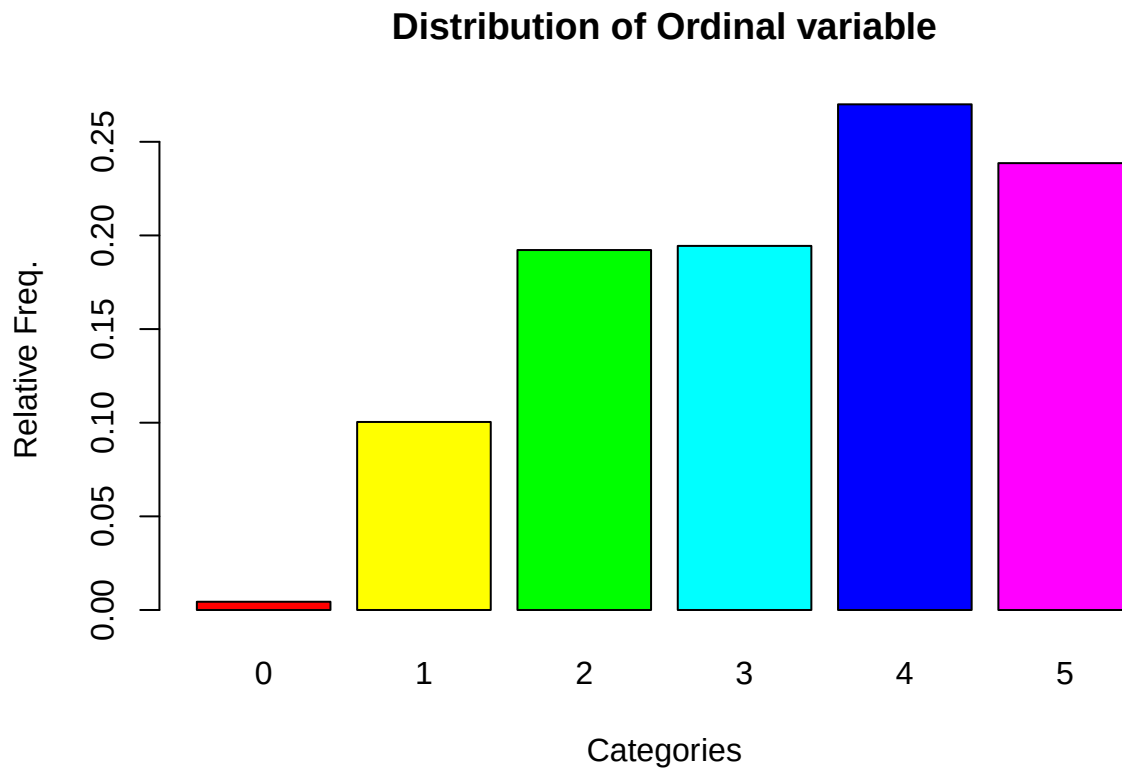
Leg.room.service is a self-describing ordinal categorical variable given by the customers in the range [0,5]. Its distribution is more uniform than previous variables but its also somewhat right skewed, with a median of 4 and mean of 3.34, deviation 1.31. There are no null values.

```
summarize_ordinal_data(train$Leg.room.service)
```

```

## [1] "Proportions: "
## data
##      0      1      2      3      4      5
## 0.0044 0.1004 0.1922 0.1944 0.2700 0.2386

```



```

## [1] "Mean: "
## [1] 3.341
## [1] "Median: "
## [1] 4
## [1] "Standard Deviation: "
## [1] 1.319798
## [1] "Number of nulls: "
## [1] 0

```

```

train$Leg.room.service <- factor(train$Leg.room.service,
                                levels = c(0, 1, 2, 3, 4, 5),
                                ordered = TRUE)

test$Leg.room.service <- factor(test$Leg.room.service,
                                levels = c(0, 1, 2, 3, 4, 5),
                                ordered = TRUE)

```

2.2.16 Variable Baggage.handling

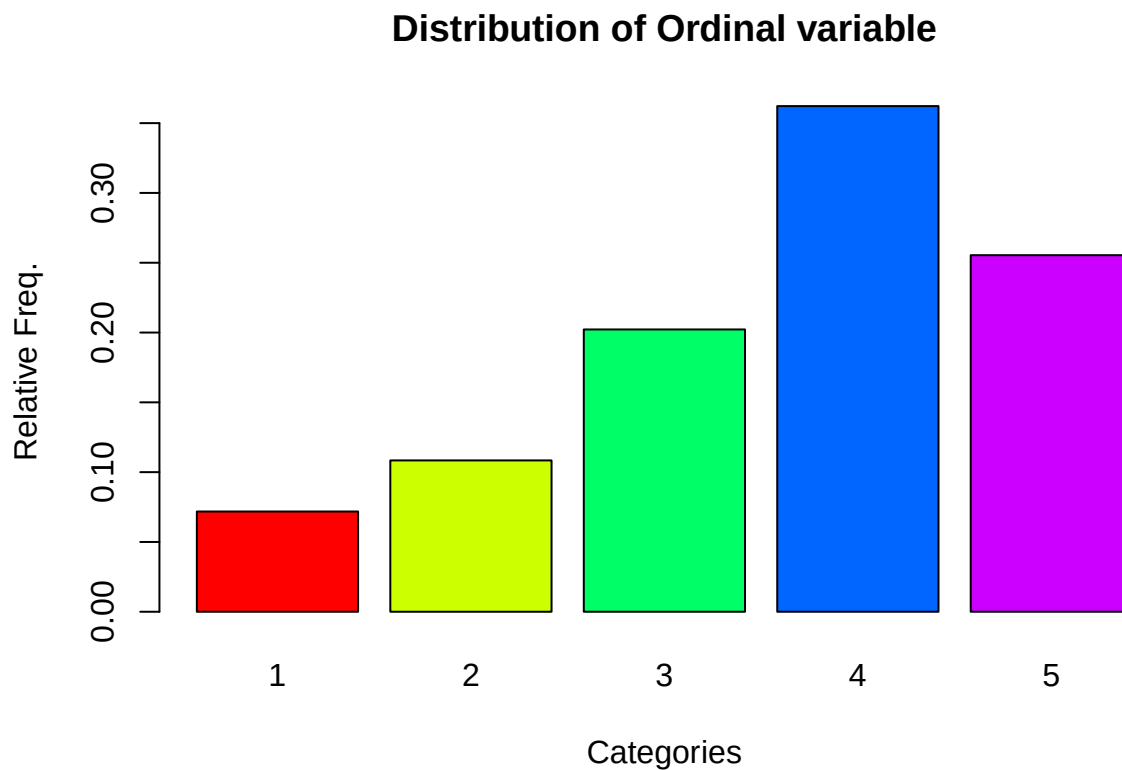
Baggage.handling is a self-describing ordinal categorical variable given by the customers in the range [1,5]. Its distribution is unimodal and highly right-skewed, having a median of 4, mean of 3.62 and small deviation of 1.18. There are no null values.

```
summarize_ordinal_data(train$Baggage.handling)
```

```

## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.0718 0.1084 0.2022 0.3622 0.2554

```



```

## [1] "Mean: "
## [1] 3.621
## [1] "Median: "
## [1] 4
## [1] "Standard Deviation: "
## [1] 1.180694

```

```
## [1] "Number of nulls: "
## [1] 0

train$Baggage.handling <- factor(train$Baggage.handling,
                                  levels = c(1, 2, 3, 4, 5),
                                  ordered = TRUE)

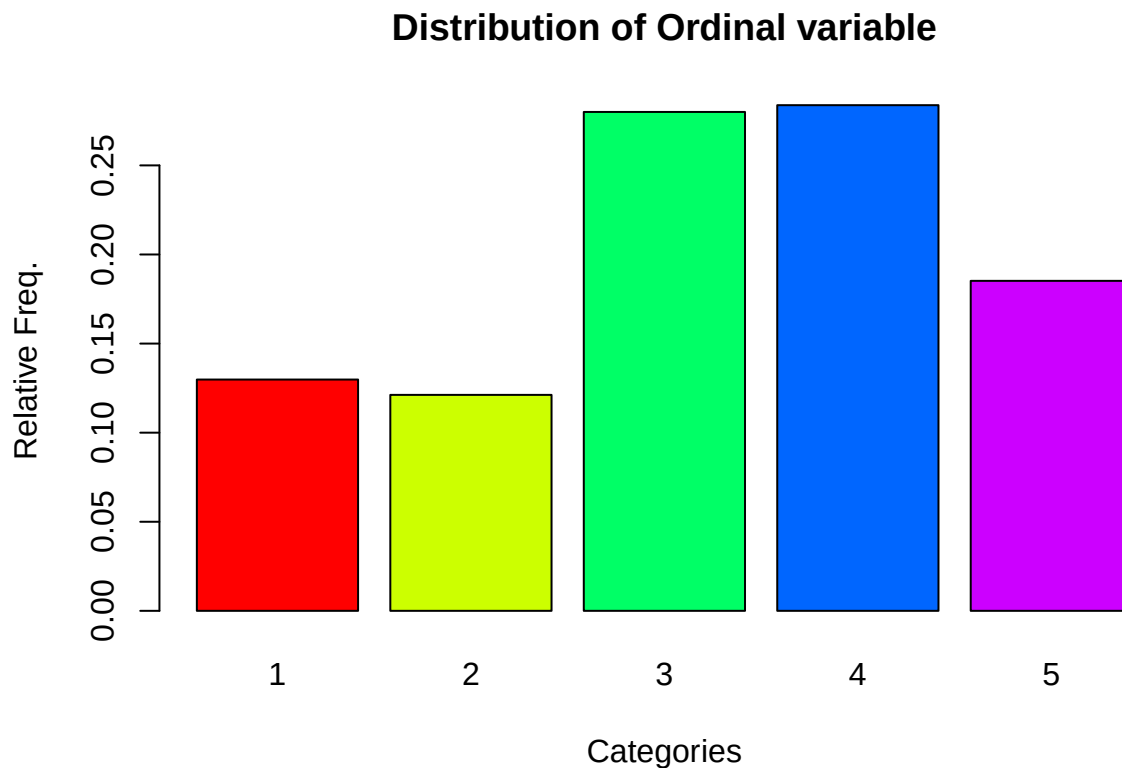
test$Baggage.handling <- factor(test$Baggage.handling,
                                 levels = c(1, 2, 3, 4, 5),
                                 ordered = TRUE)
```

2.2.17 Variable Checkin.service

Checking.service is a self-describing ordinal categorical variable given by the customers in the range [1,5]. In contrast, this distribution seems more uniform with a huge mode between 3 and 4 (median of 3, mean of 3.27, deviation 1.26). There are no null values.

```
summarize_ordinal_data(train$Checkin.service)
```

```
## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.1298 0.1212 0.2800 0.2838 0.1852
```



```
## [1] "Mean: "
## [1] 3.2734
## [1] "Median: "
## [1] 3
```

```
## [1] "Standard Deviation: "
## [1] 1.261178
## [1] "Number of nulls: "
## [1] 0

train$Checkin.service <- factor(train$Checkin.service,
                                levels = c(1, 2, 3, 4, 5),
                                ordered = TRUE)

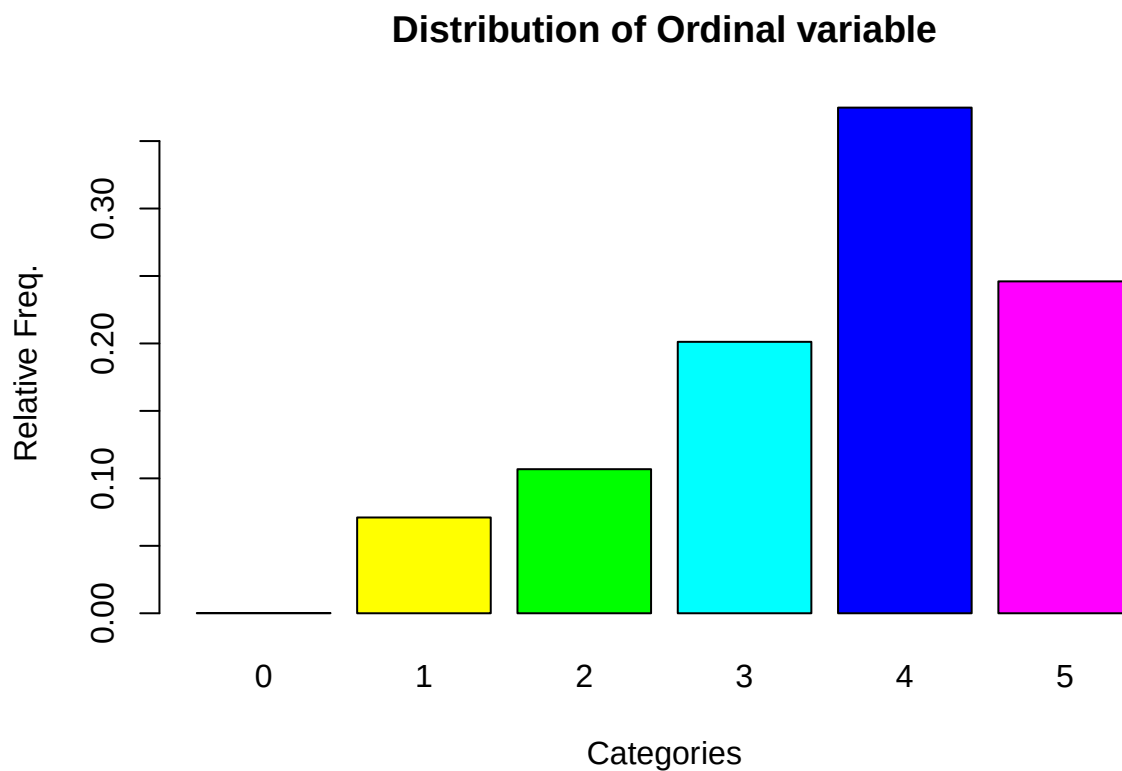
test$Checkin.service <- factor(test$Checkin.service,
                                levels = c(1, 2, 3, 4, 5),
                                ordered = TRUE)
```

2.2.18 Variable Inflight.service

Inflight.service is a self-describing ordinal categorical variable given by the customers in the range [0,5]. Its distribution is unimodal and highly right-skewed, with a median of 4, mean of 3.62 and a small deviation 1.17. There are no null values.

```
summarize_ordinal_data(train$Inflight.service)
```

```
## [1] "Proportions: "
## data
##      0      1      2      3      4      5
## 0.0002 0.0710 0.1068 0.2012 0.3748 0.2460
```



```
## [1] "Mean: "
## [1] 3.6174
```

```
## [1] "Median: "
## [1] 4
## [1] "Standard Deviation: "
## [1] 1.17068
## [1] "Number of nulls: "
## [1] 0

train$Inflight.service <- factor(train$Inflight.service,
                                levels = c(0, 1, 2, 3, 4, 5),
                                ordered = TRUE)

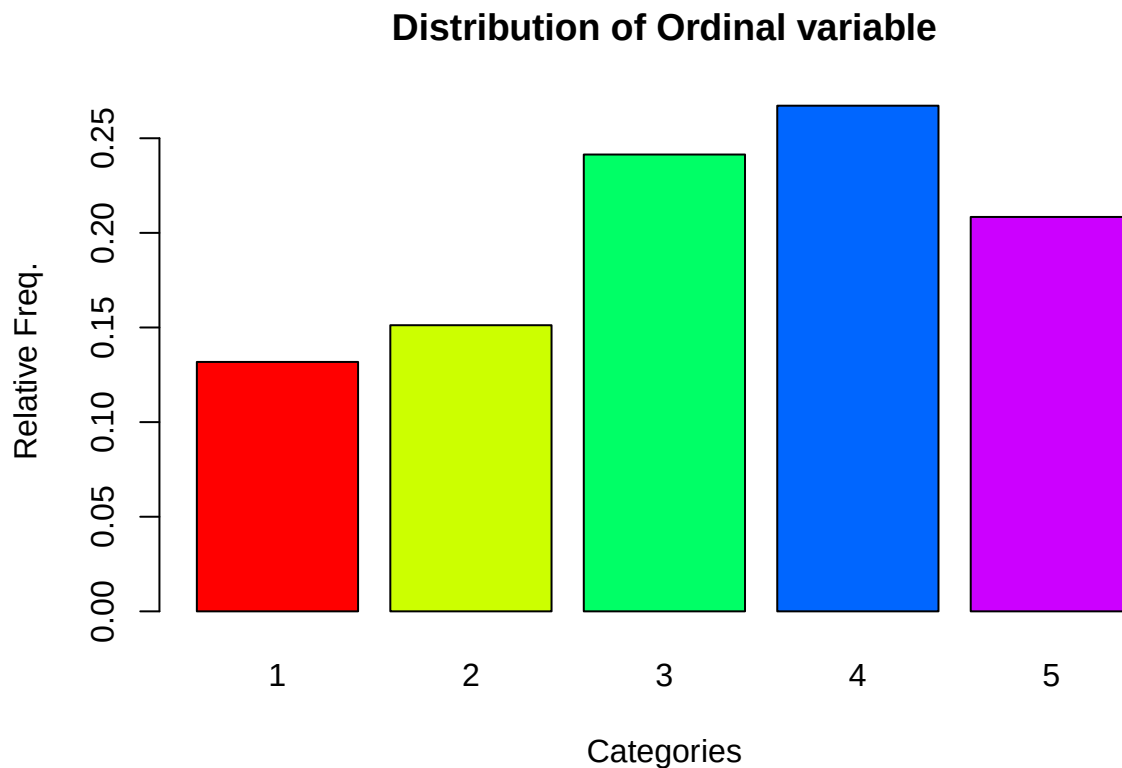
test$Inflight.service <- factor(test$Inflight.service,
                                levels = c(0, 1, 2, 3, 4, 5),
                                ordered = TRUE)
```

2.2.19 Variable Cleanliness

Cleanliness is a self-describing ordinal categorical variable given by the customers in the range [1,5]. This variable has the most uniform distribution of them all, median 3, mean 3.26 and deviation 1.3, thus skewed slightly to the right. Again, there are no null values.

```
summarize_ordinal_data(train$Cleanliness)
```

```
## [1] "Proportions: "
## data
##      1      2      3      4      5
## 0.1318 0.1512 0.2414 0.2672 0.2084
```



```
## [1] "Mean: "
## [1] 3.2692
## [1] "Median: "
## [1] 3
## [1] "Standard Deviation: "
## [1] 1.30655
## [1] "Number of nulls: "
## [1] 0

train$Cleanliness <- factor(train$Cleanliness,
                             levels = c(1, 2, 3, 4, 5),
                             ordered = TRUE)

test$Cleanliness <- factor(test$Cleanliness,
                             levels = c(1, 2, 3, 4, 5),
                             ordered = TRUE)
```

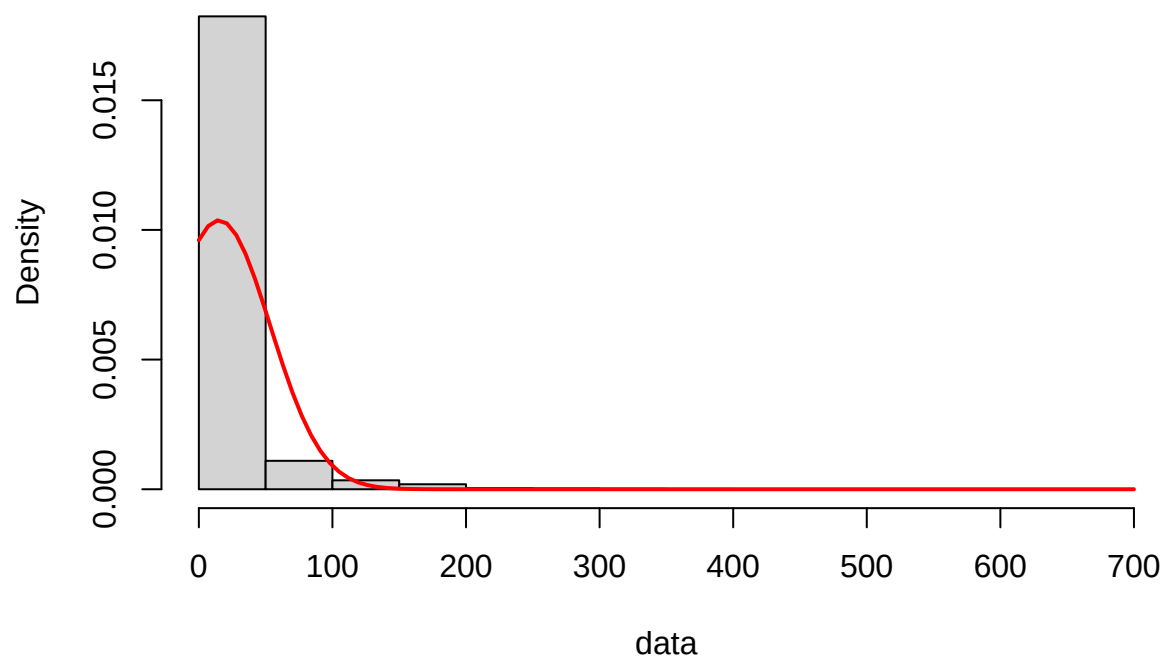
2.2.20 Variable Departure.Delay.in.Minutes

Departure.Delay.in.Minutes is a discrete numerical finite variable that represents how many minutes this rated flight has been delayed, considering its scheduled departure. Its values range from 0 to 692, with a mean of 15, median of 0 and huge deviation of 38.45. There are 675 mild outliers and 413 severe ones. Its distribution looks astonishingly exponential, with a skewness coefficient of 5.99. Both inferential tests reject the hypothesis of normality.

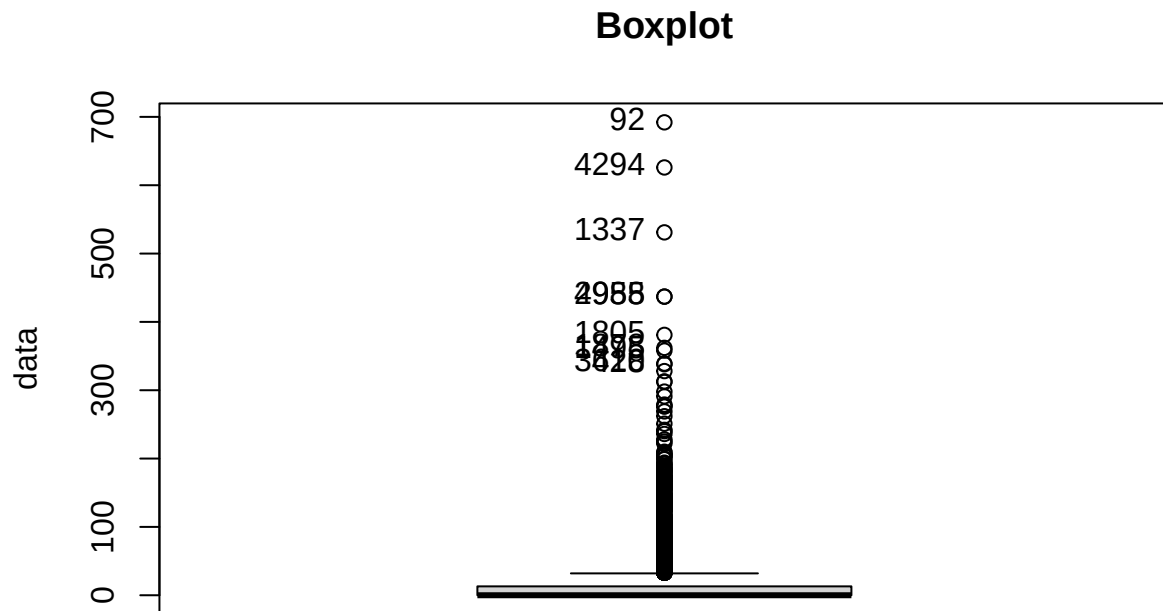
```
summary_and_normality_check(train$Departure.Delay.in.Minutes)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.00   0.00    0.00   15.08   13.00   692.00
## [1] "Mean"
## [1] 15.081
## [1] "Standard deviation"
## [1] 38.45726
```

Histogram



```
## [1] "Skewness"  
## [1] 5.995659
```



```
## [1] "Num Outliers:"
## [1] 675
## [1] "Num. severe outliers:"
## [1] 413
## [1] "Shapiro-Wilk test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.42975, p-value < 2.2e-16
##
## [1] "Kolmogorov-Smirnov test: "
##
## Warning in ks.test.default(x = data, y = "pnorm", mean = mean(data), sd =
## sd(data)): ties should not be present for the one-sample Kolmogorov-Smirnov
## test
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: data
## D = 0.34747, p-value < 2.2e-16
## alternative hypothesis: two-sided
##
## [1] 10 14 16 18 23 25 26 46 53 82 89 90 92 107 111
## [16] 117 126 128 162 168 181 183 189 206 218 224 234 245 251 263
## [31] 267 280 282 283 286 289 296 302 305 307 327 340 344 379 383
## [46] 389 406 412 423 428 447 448 461 465 470 472 496 502 506 510
```

```
## [61] 511 516 517 518 520 521 530 532 535 539 561 562 571 582 592
## [76] 595 603 605 606 624 629 631 639 646 648 649 658 663 668 670
## [91] 671 682 684 691 692 693 696 697 704 706 727 732 737 739 747
## [106] 750 755 756 768 772 777 807 808 810 813 814 817 820 824 825
## [121] 826 835 837 840 841 851 854 861 862 870 879 893 910 928 933
## [136] 935 951 960 967 971 984 1013 1014 1026 1029 1034 1038 1042 1045 1046
## [151] 1062 1069 1075 1078 1095 1129 1138 1141 1142 1143 1164 1167 1177 1179 1181
## [166] 1203 1207 1209 1210 1219 1220 1229 1231 1234 1258 1264 1266 1275 1276 1282
## [181] 1302 1329 1330 1337 1348 1351 1357 1363 1367 1369 1377 1391 1407 1409 1415
## [196] 1416 1422 1425 1430 1431 1435 1440 1446 1448 1474 1488 1495 1501 1502 1511
## [211] 1520 1529 1532 1537 1538 1543 1547 1555 1571 1579 1582 1586 1593 1606 1610
## [226] 1614 1618 1655 1665 1674 1675 1692 1696 1697 1702 1713 1737 1739 1740 1741
## [241] 1753 1758 1759 1766 1771 1801 1805 1809 1810 1836 1837 1843 1862 1865 1867
## [256] 1870 1872 1877 1878 1885 1888 1889 1890 1901 1914 1927 1941 1955 1959 1981
## [271] 1993 1994 1996 2002 2007 2016 2020 2023 2025 2041 2054 2069 2072 2089 2091
## [286] 2092 2093 2097 2101 2105 2119 2127 2134 2136 2152 2160 2167 2168 2188 2190
## [301] 2199 2203 2204 2210 2222 2255 2257 2272 2283 2290 2302 2312 2340 2341 2344
## [316] 2347 2351 2359 2360 2366 2368 2380 2381 2399 2401 2405 2412 2421 2429 2437
## [331] 2449 2450 2455 2470 2481 2510 2514 2515 2516 2531 2540 2543 2545 2546 2553
## [346] 2555 2561 2576 2577 2579 2582 2585 2618 2625 2646 2667 2672 2679 2683 2689
## [361] 2696 2720 2724 2735 2737 2743 2756 2763 2767 2780 2781 2788 2790 2791 2796
## [376] 2818 2826 2827 2842 2851 2853 2860 2862 2869 2875 2898 2913 2923 2931 2944
## [391] 2958 2974 2983 2984 2988 2992 2993 3020 3022 3028 3034 3048 3051 3075 3085
## [406] 3128 3145 3153 3158 3162 3180 3184 3204 3263 3272 3274 3282 3301 3312 3317
## [421] 3319 3320 3322 3325 3332 3334 3342 3344 3350 3354 3358 3364 3368 3374 3375
## [436] 3386 3388 3390 3394 3407 3417 3423 3428 3437 3439 3449 3456 3464 3467 3485
## [451] 3489 3493 3496 3502 3504 3506 3518 3519 3523 3524 3542 3543 3547 3555 3565
## [466] 3570 3574 3579 3587 3590 3592 3601 3610 3613 3614 3624 3630 3635 3647 3657
## [481] 3658 3660 3662 3664 3666 3667 3684 3697 3703 3705 3711 3723 3724 3729 3731
## [496] 3735 3743 3744 3757 3763 3771 3782 3789 3795 3804 3812 3813 3817 3820 3822
## [511] 3824 3829 3830 3838 3843 3845 3848 3853 3855 3868 3874 3880 3881 3887 3889
## [526] 3907 3911 3914 3915 3930 3934 3948 3951 3960 3978 3990 4005 4006 4010 4023
## [541] 4028 4033 4053 4055 4062 4070 4074 4076 4081 4091 4100 4108 4109 4117 4120
## [556] 4132 4134 4135 4139 4141 4149 4155 4165 4174 4177 4184 4188 4192 4206 4216
## [571] 4225 4228 4230 4234 4236 4238 4254 4255 4258 4259 4260 4267 4272 4276 4279
## [586] 4280 4294 4318 4332 4340 4352 4353 4355 4363 4365 4367 4401 4409 4419 4422
## [601] 4425 4454 4463 4472 4488 4502 4518 4532 4546 4548 4550 4561 4575 4583 4584
## [616] 4591 4596 4602 4604 4608 4617 4627 4645 4651 4652 4673 4683 4691 4697 4705
## [631] 4710 4713 4714 4716 4717 4719 4722 4723 4728 4738 4740 4750 4752 4755 4758
## [646] 4767 4775 4789 4791 4798 4820 4821 4840 4845 4850 4857 4858 4876 4891 4898
## [661] 4915 4916 4918 4920 4929 4931 4933 4937 4946 4955 4965 4967 4982 4990 4991
```

2.2.21 Variable Arrival.Delay.in.Minutes

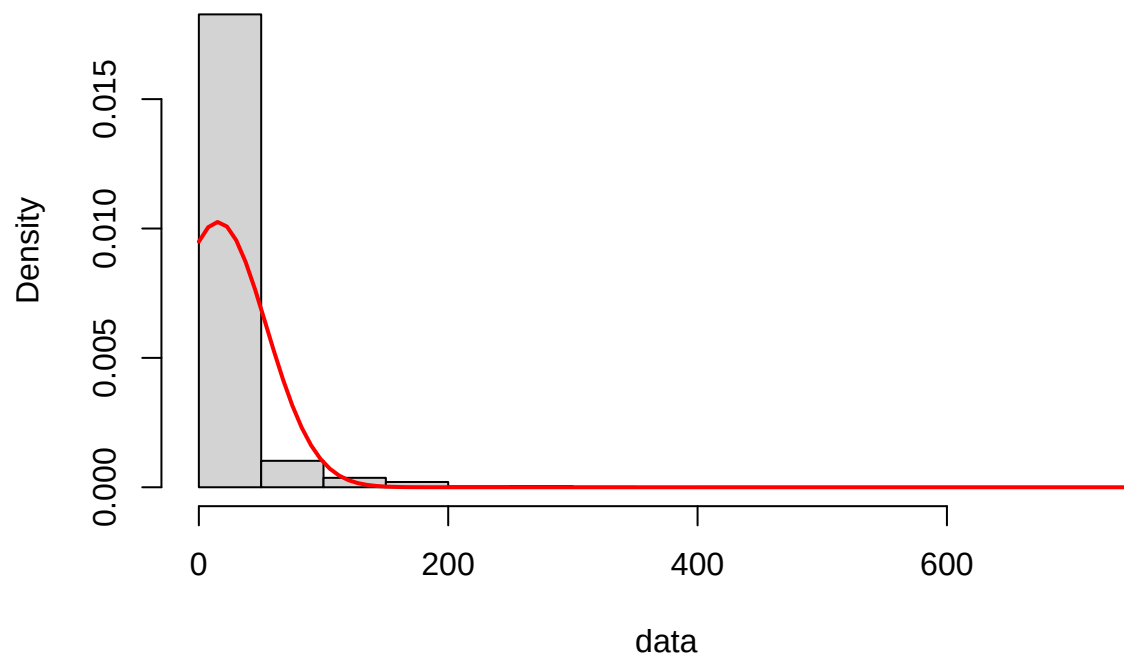
Arrival.Delay.in.minutes is a discrete numerical finite variable that represents how many minutes this rated flight has been delayed, considering its scheduled arrival. Similarly to departure delay, its distribution has an astonishingly exponential shape, and a huge coefficient of skewness of 6.06. Has a mean of 15.28 but a median of 0, with values ranging between [0, 702] minutes, 12 null values, 669 mild outliers and 410 severe ones. Both inferential tests reject the hypothesis of normality.

```
summary_and_normality_check(train$Arrival.Delay.in.Minutes)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##      0.00   0.00    0.00   15.28   13.00   702.00     12
```

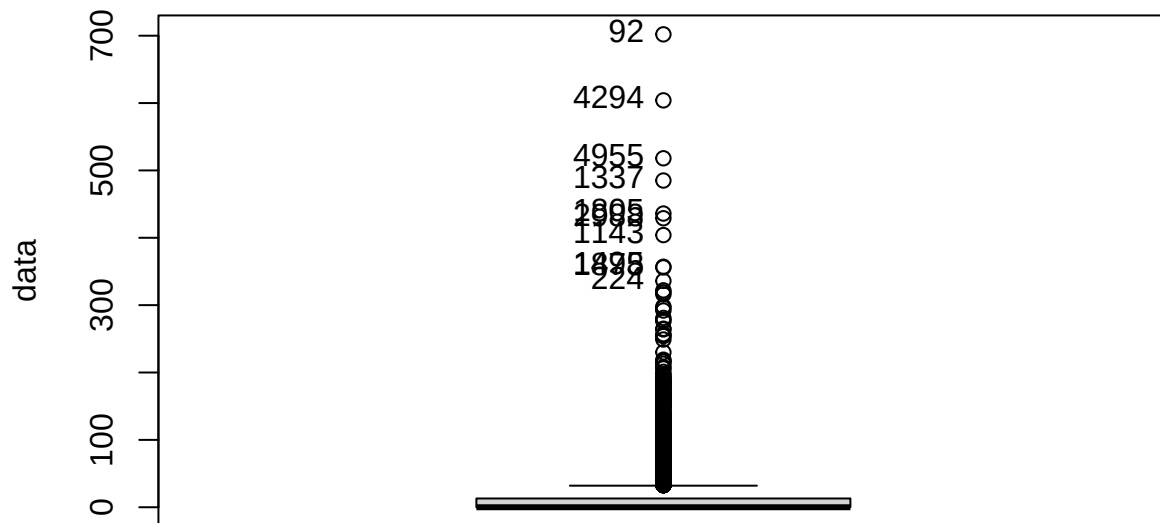
```
## [1] "Mean"  
## [1] 15.27506  
## [1] "Standard deviation"  
## [1] 38.91399
```

Histogram



```
## [1] "Skewness"  
## [1] 6.060705
```

Boxplot



```
## [1] "Num Outliers:"
## [1] 669
## [1] "Num. severe outliers:"
## [1] 410
## [1] "Shapiro-Wilk test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.42739, p-value < 2.2e-16
##
## [1] "Kolmogorov-Smirnov test: "
##
## Warning in ks.test.default(x = data, y = "pnorm", mean = mean(data), sd =
## sd(data)): ties should not be present for the one-sample Kolmogorov-Smirnov
## test
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: data
## D = NA, p-value = NA
## alternative hypothesis: two-sided
##
## [1] 9 14 18 23 25 26 46 53 82 89 90 92 107 111 117
## [16] 126 128 162 168 181 183 189 206 218 224 227 234 251 267 280
## [31] 283 286 289 302 327 330 344 349 361 368 379 383 406 412 423
## [46] 428 438 447 448 461 464 465 470 472 496 502 510 511 516 517
```

```
## [61] 518 530 535 539 561 562 571 582 592 595 603 605 618 619 624
## [76] 629 633 636 639 644 646 648 649 663 668 670 671 682 691 692
## [91] 693 696 697 702 704 706 727 737 739 747 750 755 756 768 775
## [106] 777 807 808 810 813 814 817 820 824 825 826 834 835 840 841
## [121] 861 862 870 879 893 898 906 910 928 933 935 951 960 967 971
## [136] 984 1002 1013 1014 1026 1029 1034 1038 1042 1045 1046 1062 1069 1078 1084
## [151] 1105 1119 1129 1138 1139 1141 1142 1143 1164 1167 1181 1203 1207 1209 1210
## [166] 1220 1231 1234 1258 1264 1266 1274 1275 1294 1302 1329 1330 1337 1351 1357
## [181] 1363 1367 1369 1377 1379 1388 1391 1407 1409 1415 1416 1425 1430 1431 1435
## [196] 1439 1440 1445 1448 1449 1474 1486 1488 1495 1501 1502 1511 1520 1529 1532
## [211] 1533 1537 1538 1543 1548 1579 1586 1594 1610 1618 1627 1655 1674 1675 1696
## [226] 1697 1702 1713 1737 1739 1740 1758 1766 1771 1801 1805 1824 1836 1837 1843
## [241] 1862 1865 1867 1869 1870 1872 1877 1878 1888 1889 1890 1893 1901 1927 1941
## [256] 1955 1959 1981 1987 1989 1993 1996 2002 2007 2016 2020 2025 2041 2054 2068
## [271] 2069 2092 2093 2097 2105 2108 2119 2127 2134 2136 2152 2160 2167 2168 2178
## [286] 2185 2188 2190 2203 2204 2210 2222 2255 2257 2272 2280 2283 2290 2302 2312
## [301] 2341 2344 2347 2351 2354 2360 2366 2368 2380 2381 2399 2401 2429 2437 2449
## [316] 2455 2469 2470 2481 2496 2510 2514 2515 2516 2539 2540 2545 2546 2553 2555
## [331] 2561 2577 2579 2582 2585 2598 2600 2609 2618 2625 2647 2678 2679 2683 2689
## [346] 2696 2700 2709 2716 2720 2724 2735 2737 2743 2763 2780 2781 2788 2790 2791
## [361] 2796 2818 2825 2826 2827 2842 2851 2860 2862 2864 2869 2870 2875 2881 2889
## [376] 2898 2913 2931 2947 2954 2958 2962 2974 2976 2983 2984 2988 2992 2993 3007
## [391] 3020 3022 3028 3034 3047 3048 3050 3051 3075 3085 3126 3128 3145 3153 3158
## [406] 3160 3173 3182 3184 3203 3204 3247 3263 3272 3274 3277 3282 3301 3312 3319
## [421] 3320 3322 3332 3336 3342 3344 3350 3354 3364 3368 3374 3375 3386 3390 3394
## [436] 3407 3423 3427 3428 3439 3449 3464 3466 3467 3489 3496 3502 3504 3506 3518
## [451] 3519 3523 3524 3542 3543 3547 3553 3555 3565 3570 3574 3579 3587 3590 3592
## [466] 3601 3610 3613 3614 3620 3622 3624 3630 3635 3647 3657 3658 3660 3662 3664
## [481] 3667 3684 3697 3703 3705 3709 3711 3722 3723 3724 3729 3731 3735 3743 3763
## [496] 3769 3771 3782 3789 3790 3795 3804 3812 3813 3817 3820 3822 3829 3838 3845
## [511] 3848 3853 3855 3856 3868 3874 3880 3881 3887 3889 3907 3911 3914 3915 3930
## [526] 3934 3937 3938 3948 3960 4000 4005 4006 4010 4023 4028 4053 4055 4062 4070
## [541] 4074 4075 4076 4081 4090 4091 4092 4100 4105 4108 4109 4117 4120 4132 4134
## [556] 4135 4141 4149 4155 4165 4174 4177 4188 4192 4195 4205 4206 4211 4216 4225
## [571] 4228 4230 4236 4255 4258 4260 4261 4267 4274 4276 4279 4294 4318 4332 4348
## [586] 4352 4353 4355 4365 4367 4380 4401 4409 4413 4419 4422 4425 4454 4463 4472
## [601] 4486 4488 4502 4518 4532 4546 4548 4550 4561 4583 4584 4591 4596 4602 4604
## [616] 4608 4625 4642 4645 4651 4652 4673 4683 4691 4697 4705 4710 4713 4714 4716
## [631] 4717 4719 4722 4723 4728 4731 4738 4740 4745 4750 4752 4755 4767 4775 4820
## [646] 4821 4832 4840 4843 4845 4850 4857 4858 4891 4898 4908 4909 4918 4920 4931
## [661] 4933 4937 4946 4955 4965 4967 4982 4990 4991
```

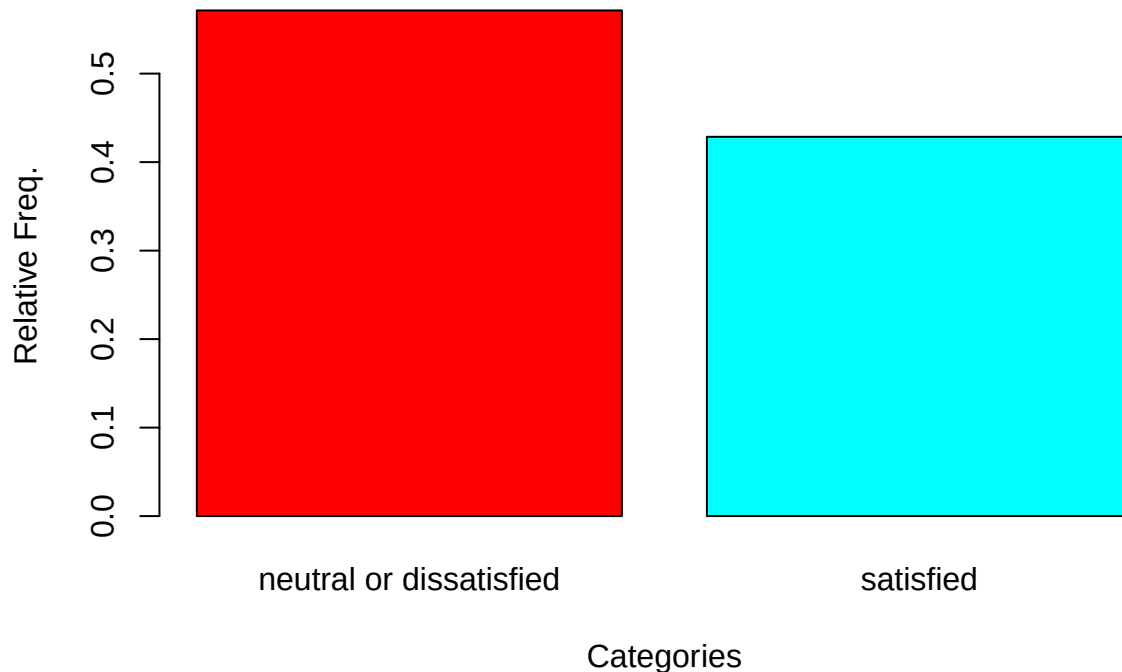
2.2.22 Variable satisfaction

satisfaction is a target binary categorical variable that represents whether the customer is satisfied or not/neutral with its overall experience. The dataset is somewhat balanced with a slight bias towards neutral/dissatisfied (57.14% vs 42.86%). There are no null values.

```
#Transform to factor
summarize_ordinal_data(train$satisfaction)
```

```
## [1] "Proportions: "
## data
## neutral or dissatisfied      satisfied
##                0.5714                0.4286
```

Distribution of Ordinal variable



```
## [1] "Mean: "
## Warning in mean.default(data): argument is not numeric or logical: returning NA
## [1] NA
## [1] "Median: "
## Warning in mean.default(sort(x, partial = half + 0L:1L)[half + 0L:1L]):
## argument is not numeric or logical: returning NA
## [1] NA
## [1] "Standard Deviation: "
## Warning in var(if (is.vector(x) || is.factor(x)) x else as.double(x), na.rm =
## na.rm): NAs introduced by coercion
## [1] NA
## [1] "Number of nulls: "
## [1] 0
train$satisfaction <- as.factor(train$satisfaction)
test$satisfaction <- as.factor(test$satisfaction)
```

2.2.23 New variables defintion

TODO: mirar si això ens importa o no realment, per modelling pot ser interesant.

```
summary(train)
```

```
##          X          id          Gender          Customer.Type
## Min.    :    4  Min.    :   21  Female:2548  disloyal Customer: 934
```

```

## 1st Qu.: 26305    1st Qu.: 33729    Male :2452    Loyal Customer :4066
## Median : 51833    Median : 64916
## Mean : 51963      Mean : 65426
## 3rd Qu.: 78259    3rd Qu.: 97538
## Max. :103866      Max. :129864
##
##      Age              Type.of.Travel      Class      Flight.Distance
## Min. : 7.00    Business travel:3460    Business:2361    Min. : 67
## 1st Qu.:27.00    Personal Travel:1540    Eco :2278    1st Qu.: 414
## Median :40.00                                Eco Plus: 361    Median : 842
## Mean :39.41                                           Mean :1194
## 3rd Qu.:51.00                                           3rd Qu.:1754
## Max. :85.00                                           Max. :4983
##
## Inflight.wifi.service Departure.Arrival.time.convenient Ease.of.Online.booking
## 0: 149              0: 247              0: 217
## 1: 878              1: 752              1: 851
## 2:1209              2: 801              2:1093
## 3:1242              3: 879              3:1203
## 4: 974              4:1250              4: 970
## 5: 548              5:1071              5: 666
##
## Gate.location Food.and.drink Online.boarding Seat.comfort
## 1: 837            0: 5              0: 119            0: 0
## 2: 909            1: 669            1: 536            1: 627
## 3:1380            2: 985            2: 815            2: 704
## 4:1188            3:1106            3:1030            3: 879
## 5: 686            4:1186            4:1538            4:1546
##                  5:1049            5: 962            5:1244
##
## Inflight.entertainment On.board.service Leg.room.service Baggage.handling
## Min. :0.000            0: 1              0: 22             1: 359
## 1st Qu.:2.000            1: 556            1: 502            2: 542
## Median :4.000            2: 713            2: 961            3:1011
## Mean :3.343            3:1129            3: 972            4:1811
## 3rd Qu.:4.000            4:1535            4:1350            5:1277
## Max. :5.000            5:1066            5:1193
##
## Checkin.service Inflight.service Cleanliness Departure.Delay.in.Minutes
## 1: 649            0: 1              1: 659            Min. : 0.00
## 2: 606            1: 355            2: 756            1st Qu.: 0.00
## 3:1400            2: 534            3:1207            Median : 0.00
## 4:1419            3:1006            4:1336            Mean : 15.08
## 5: 926            4:1874            5:1042            3rd Qu.: 13.00
##                  5:1230                                Max. :692.00
##
## Arrival.Delay.in.Minutes              satisfaction
## Min. : 0.00          neutral or dissatisfied:2857
## 1st Qu.: 0.00          satisfied :2143
## Median : 0.00
## Mean : 15.28
## 3rd Qu.: 13.00
## Max. :702.00
## NA's :12

```

2.3 Data quality report

This section provides a comprehensive overview of the dataset's quality, complementing the prior identification of univariate outliers and missing values with a visual summary to highlight key patterns and anomalies. The data quality report is structured into two parts: an analysis regarding the variables and a review of individual records (stating the number of outliers per individual and identifying multivariate outliers).

2.3.1 Variables

2.3.1.1 Missing values and Errors

2.3.1.2 Outliers

2.3.2 Individuals

2.3.2.1 Missing values and Errors

2.3.2.2 Outliers

2.3.2.3 Multivariant Outliers

2.3.3 Data Quality Relationships

Here we make variables for multivariate outliers and stuff ...

2.4 Imputation

In this section, all the missing values previously identified are addressed through an imputation process. To avoid repetitive code, a function is first defined to perform the imputation consistently across variables. The imputation is then carried out for each variable with missing values, and a brief commentary is provided for each to discuss the imputation approach and its impact on the data.

Note that values are imputed both in the training and test sets, as stated in the statement of the project.

```
imputation_function <- function(variable) {

  num_vars <- df[sapply(df, is.numeric)]
  miss_ids <- which(is.na(df[[variable]]))

  # Perform PCA-based imputation
  res_imp <- imputePCA(num_vars)
  imp_values <- res_imp$completeObs[, variable]

  # Calculate quantiles for validation
  quantiles_df <- quantile(df[[variable]], probs = seq(0, 1, 0.1), na.rm = TRUE)
  quantiles_imp <- quantile(imp_values, probs = seq(0, 1, 0.1), na.rm = FALSE)

  # Plot the quantile comparison
  plot(seq(0, 1, 0.1), quantiles_df, type = "l", col = "blue", lwd = 2,
        xlab = "Quantile", ylab = "Value",
        main = paste("Quantile Comparison for", variable))
  lines(seq(0, 1, 0.1), quantiles_imp, col = "red", lwd = 2)
  legend("topright", legend = c(variable, paste("imp_", variable, sep = "")),
        col = c("blue", "red"), lwd = 2)

  return(list(
```

```

    na_indexes = miss_ids,
    imp_values = imp_values[miss_ids]
  ))
}

imputation_function_test <- function(variable) {

  num_vars <- test[sapply(test, is.numeric)]
  miss_ids <- which(is.na(test[[variable]]))

  # Perform PCA-based imputation
  res_imp <- imputePCA(num_vars)
  imp_values <- res_imp$completeObs[, variable]

  # Calculate quantiles for validation
  quantiles_df <- quantile(test[[variable]],
                           probs = seq(0, 1, 0.1), na.rm = TRUE)
  quantiles_imp <- quantile(imp_values, probs = seq(0, 1, 0.1),
                           na.rm = FALSE)

  # Plot the quantile comparison
  plot(seq(0, 1, 0.1), quantiles_df, type = "l", col = "blue", lwd = 2,
       xlab = "Quantile", ylab = "Value",
       main = paste("Quantile Comparison for", variable))
  lines(seq(0, 1, 0.1), quantiles_imp, col = "red", lwd = 2)
  legend("topright", legend = c(variable, paste("imp_", variable, sep = "")),
       col = c("blue", "red"), lwd = 2)

  return(list(
    na_indexes = miss_ids,
    imp_values = imp_values[miss_ids]
  ))
}

```

2.4.1 Arrival.Delay.in.Minutes

2.5 Profiling analysis

The target binary variable **satisfaction** splits the data unevenly: 57.14% neutral or dissatisfied vs 42.86% satisfied. Moreover this variable contains no null values at all. To delve deeper into the main differences between these 2 groups, a clustering profiling will be performed.

```
res_cat <- catdes(train, proba=0.05, num.var=25)
```

```
res_cat$test.chi2
```

##	p.value	df
## Online.boarding	0.000000e+00	5
## Inflight.wifi.service	3.032475e-299	5
## Class	3.762928e-290	2
## Type.of.Travel	3.141061e-216	1
## Seat.comfort	2.391451e-160	4
## On.board.service	3.112166e-122	5

```
## Leg.room.service          1.522328e-118  5
## Ease.of.Online.booking    6.281922e-104  5
## Cleanliness               1.661592e-102  4
## Baggage.handling          4.589485e-87  4
## Inflight.service          7.601288e-82  5
## Checkin.service           1.310091e-59  4
## Food.and.drink            5.817625e-48  5
## Customer.Type             1.626909e-46  1
## Gate.location             2.329562e-25  4
## Departure.Arrival.time.convenient 1.538508e-02  5
```

```
res_cat$quanti.var
```

```
##              Eta2      P-value
## Inflight.entertainment 0.160201642 9.185675e-192
## Flight.Distance        0.090411953 5.327880e-105
## Age                    0.021854338 7.895448e-26
## Arrival.Delay.in.Minutes 0.001750924 3.118368e-03
## Departure.Delay.in.Minutes 0.001095035 1.928555e-02
```

```
res_cat$quanti
```

```
## $`neutral or dissatisfied`
```

```
##              v.test Mean in category Overall mean
## Arrival.Delay.in.Minutes 2.957488      16.684672      15.27506
## Departure.Delay.in.Minutes 2.339675      16.183059      15.08100
## Age -10.452265      37.502275      39.41260
## Flight.Distance -21.259571      934.005950      1194.20540
## Inflight.entertainment -28.299258      2.880994      3.34260
##              sd in category Overall sd      p.value
## Arrival.Delay.in.Minutes 41.514521      38.910088      3.101571e-03
## Departure.Delay.in.Minutes 40.886883      38.453416      1.930052e-02
## Age 16.348928      14.920468      1.430676e-25
## Flight.Distance 809.057576      999.165622      2.688635e-100
## Inflight.entertainment 1.320994      1.331625      3.529307e-176
```

```
## $satisfied
```

```
##              v.test Mean in category Overall mean
## Inflight.entertainment 28.299258      3.958003      3.34260
## Flight.Distance 21.259571      1541.097527      1194.20540
## Age 10.452265      41.959403      39.41260
## Departure.Delay.in.Minutes -2.339675      13.611759      15.08100
## Arrival.Delay.in.Minutes -2.959563      13.394478      15.27506
##              sd in category Overall sd      p.value
## Inflight.entertainment 1.07146      1.331625      3.529307e-176
## Flight.Distance 1116.25574      999.165622      2.688635e-100
## Age 12.31746      14.920468      1.430676e-25
## Departure.Delay.in.Minutes 34.89252      38.453416      1.930052e-02
## Arrival.Delay.in.Minutes 35.04801      38.910088      3.080758e-03
```

3. Modeling

3.1 Model Discovery and Feature Selection

3.2 Effects study

3.3 Residual analysis and Influential data study

3.4 Effects study

3.5 Model evaluation

Goodness of fit a la lidia li encanta aquesta paraula!!!!

4 Conclusion and Further Work

Extensissimo prolongado relato